

MCSA-P01

New Office Infrastructure — complete implementation workbook

Independent project; foundation and specialist instructions with diagrams at the point of use

Windows Server 2016 training lab

Revised edition 2.0 | 16 September 2026

Prepared for Harish and the training team

One implementation workbook: prerequisites, diagrams, tasks, checks and handover.

Trainee / team _____

Batch _____ Start date _____

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SECTION 0 PROJECT CHARTER

Field	Definition
Company	Aster Business Services; fictional Pune services organisation; 60 staff; growth plan to 90; one actual lab site
Project	MCSA-P01 New Office Infrastructure
Tier	Foundation
Effort	30 hours for guided specialist work, practice, drills and evidence; Projects 2-6 add 20-30 hours if rebuilding the included foundation from zero
In scope	AD DS, DNS, DHCP, SMB and NTFS, Group Policy, Windows Server Backup, Firewall and event logs
Out of scope	Paid employment claims, live company data, production rollout, public exposure, enterprise performance/load guarantees and technologies not explicitly implemented
Prerequisite	Foundation B01-B16 accepted or built using the included full guide
Success	All task verifications and five drills pass with real evidence; evaluator rubric and sign-off completed

Business scenario: Anita in HR cannot retrieve onboarding files when a colleague is absent. Rohan in Finance can browse folders outside his role. Meera accidentally deletes an operating procedure. Aster needs named accounts, controlled access and a recovery process that works.

Skill family	Historical MCSA objective family	CPR evidence	Where demonstrated
AD DS, DNS, DHCP, SMB and NTFS, Group Policy, Windows Server Backup, Firewall and event logs	70-740 installation/storage; 70-741 networking; 70-742 identity, as applicable	CPR Tracker recorded days 1, 2, 3, 4, 7, 8, 9, 10, 15, 16, 21, 22	Task verification and Section 7 tests
Shared foundation	AD DS, DNS, DHCP, policy and storage fundamentals	Days 1-4, 7-10 and revision	Included B01-B16 guide

The historical exam mapping is at objective-family level, not a certification award or complete examination mapping. Retired-track course attendance must not be represented as holding an earned Microsoft certification.

Phase	Order and effort allowance
Preparation	Validate foundation and read concepts/diagrams: 20 percent
Implementation	Complete numbered project tasks: 40 percent
Validation	Functional checks, failures and recovery: 25 percent
Handover	Evidence, report and technical defence: 15 percent

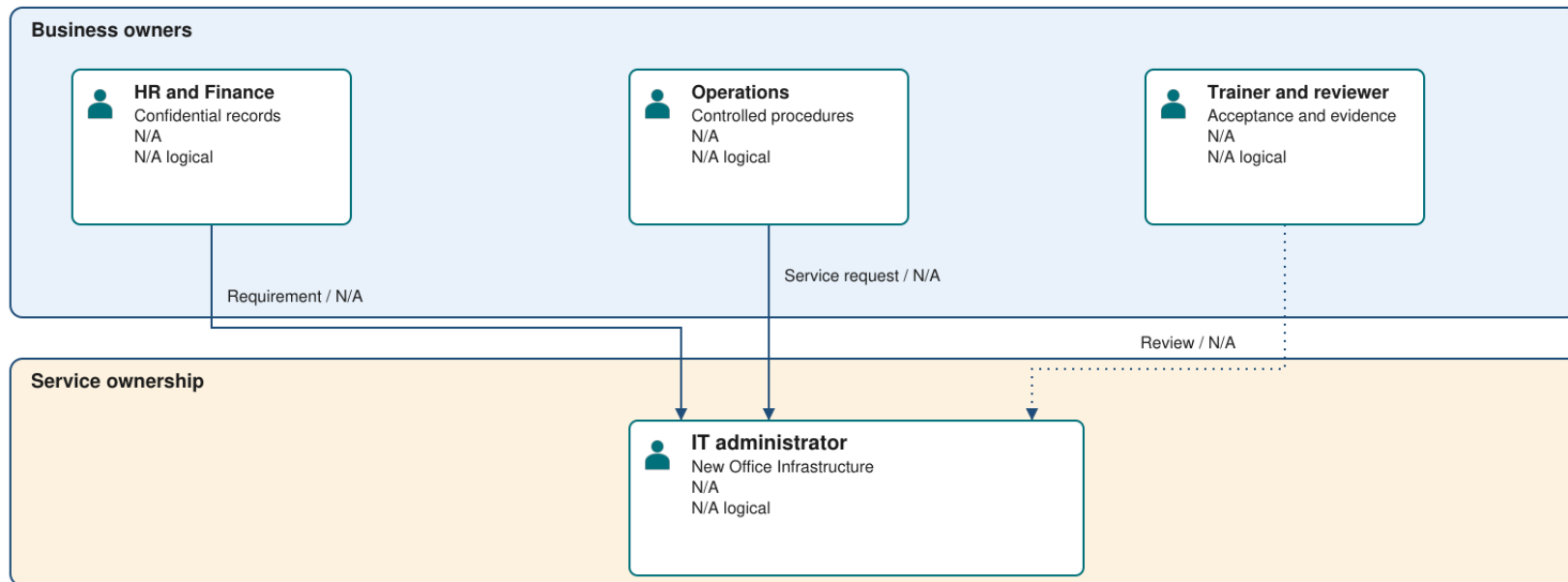
Resume outcomes after successful execution: built the documented service; proved allowed and denied behaviour; recovered from five controlled faults; produced a repeatable operations handover. Use the precise wording and honesty guidance in Section 11.

Edition 2.0 consolidates the prerequisite build into this workbook and embeds diagrams at their implementation phases. Existing task and evidence IDs remain stable. There is one ZIP per project; multipart downloads are unnecessary.

D1 | Introduction: the business service this project delivers

MCSA-P01 | D1 | Business context and ownership

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Private lab addresses | Open the SVG at full size for readable labels

Legend solid: traffic dashed: replication
dotted: management red: recovery N/A: logical

Business context and ownership

Current phase: Introduction: the business service this project delivers. Start with the people who experience the problem. The arrows describe responsibility and requirements, not network packets. IT must return working evidence to these stakeholders. Logical nodes have no operating-system address.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

SECTION 1 WHY

1.1 The business pain

Anita in HR cannot retrieve onboarding files when a colleague is absent. Rohan in Finance can browse folders outside his role. Meera accidentally deletes an operating procedure. Aster needs named accounts, controlled access and a recovery process that works.

Pain	Affected role	Business impact	Illustrative cost	Risk
Work interrupted by unavailable service	Department user	Delayed work and repeat support calls	4 people x INR 250/hour = INR 1,000/hour; fictional teaching assumption	High
Uncontrolled access or configuration	Data owner and IT	Unexplained access and inconsistent service	No credible incident-loss figure assigned	High
No recovery or handover evidence	IT lead	Longer diagnosis and uncertain restoration	2 technicians x INR 400/hour = INR 800/hour; fictional assumption	Medium

If nothing changes for 12 months, the fictional growth from 60 to 90 employees increases manual work and the number of inconsistent endpoints. These assumptions explain design choices; the lab does not validate a 90-user workload or quantify savings.

Business driver	Measurable completion threshold
Reliable service	Every task-specific verification passes with actual output
Controlled access	All stated positive/negative permission and connectivity cases match
Recoverability	Every drill restored to the accepted state; backup projects prove hash-matched restore
Operational clarity	Another trainee can follow the handover and reproduce three selected checks
Job readiness	Trainee explains the why, demonstrates a small live change and records a truthful lab project story

SECTION 2 WHAT

The foundational and specialist concepts are included below. Read them before the Section 5 build.

AD DS

Problem solved: Centralise user and computer identity.

Everyday analogy: A central staff register.

How it works: AD stores directory objects on a domain controller. Clients locate a DC through DNS and use domain authentication. Computer membership creates a trust relationship.

Term group	Meaning in this project
Forest; domain; DC; OU; SID	ad.aster.example on AST-DC01
Failure consequence	Wrong DNS or broken trust prevents fresh authentication and policy processing

DNS

Problem solved: Resolve server names and locate AD services.

Everyday analogy: A directory with addresses and service listings.

How it works: A/AAAA records locate hosts; SRV records locate services. AD-integrated DNS data is stored in directory partitions. DNS resolution is a dependency, not proof that a service port is open.

Term group	Meaning in this project
Zone; A; PTR; SRV; FQDN	FS01 A record .20 and LDAP DC-locator records
Failure consequence	Joining the domain fails even when ping by IP works

DHCP

Problem solved: Assign consistent client network settings.

Everyday analogy: A reception desk allocating temporary desks.

How it works: Discover, Offer, Request and Acknowledge establish a lease. Options carry DNS/domain settings. Infrastructure addresses remain outside the client pool.

Term group	Meaning in this project
Scope; lease; exclusion; option 006; authorisation	192.168.50.100-149; DNS .10
Failure consequence	APIPA or wrong DNS causes connectivity and domain failures

SMB and NTFS

Problem solved: Share files while enforcing departmental access.

Everyday analogy: A building access check followed by a room access check.

How it works: Remote file operations must satisfy both share and NTFS permissions. Account tokens include group memberships; a new logon may be required after membership changes.

Term group	Meaning in this project
UNC; share ACL; NTFS ACL; inheritance; AGDLP	HR\$ and DL_FS_HR_Modify
Failure consequence	Wrong grants cause either data exposure or legitimate access denial

Group Policy

Problem solved: Apply repeatable settings to the correct identities.

Everyday analogy: Instructions delivered to the right department.

How it works: GPO scope, links, filtering and inheritance determine applicability. User policies follow user objects; computer policies follow computer objects. Domain password policy has domain scope.

Term group	Meaning in this project
GPO; link; user scope; computer scope; resultant policy	HR Control Panel restriction

Term group	Meaning in this project
Failure consequence	A created but unlinked policy has no effect

Windows Server Backup

Problem solved: Recover a prior file state.

Everyday analogy: A dated archive with a retrieval procedure.

How it works: WSB creates a recoverable backup using Windows backup/VSS facilities. Restore the selected version to another location and compare content. Reusing a remote destination may overwrite the previous backup.

Term group	Meaning in this project
Recovery point; RPO; RTO; VSS; restore	FS01 data to BK01 restricted share
Failure consequence	A job success message alone cannot prove restore integrity

Firewall and event logs

Problem solved: Control connectivity and explain failures.

Everyday analogy: A gatekeeper and incident diary.

How it works: Rules match profiles, directions, protocols and addresses. Logs record component events; timestamps and providers help connect a symptom to a cause.

Term group	Meaning in this project
Profile; inbound; port; provider; event ID	SMB 445; Security/System/Application logs
Failure consequence	Disabling all firewall profiles hides the fault and changes the security boundary

Design decision register

Decision	Options	Choice	Why	Trade-off
Recovery target	Copy; local backup; remote backup	Remote WSB file backup	Separate logical target and an actual restoration	Both VMs share one host; host loss is not covered
Retention	Overwrite one destination; dated destinations	Two dated test destinations	Allows the trainee to identify a recovery point	Capacity and cleanup must be managed

Common decisions: one forest ad.aster.example with NetBIOS ASTER; static infrastructure addresses outside DHCP; user OUs for policy and security groups for access; Server 2016 Desktop Experience; no paid third-party tooling. The foundation begins with all five FSMO roles on DC01; only Project 4 transfers and restores the PDC role.

Standard	Exact convention
Servers	AST-DC01 .10; AST-FS01 .20; AST-BK01 .30; specialist values in Section 4
Accounts	anita.hr, vikas.hr; rohan.fin, neha.fin; meera.ops, arjun.ops
OUs	Aster > Users > HR / Finance / Operations; Groups; Servers; Workstations; AdminAccounts
Permission groups	GG_<Department>_Users nested in DL_FS_<Department>_Modify
Department shares	HR\$, Finance\$, Operations\$ under D:\Departments
Policy	HR user GPO linked to HR OU; domain password/lockout at domain scope

Foundation concept: AD DS

Problem solved: Centralise user and computer identity. Everyday analogy: A central staff register.

AD stores directory objects on a domain controller. Clients locate a DC through DNS and use domain authentication. Computer membership creates a trust relationship.

Terms	Example in this build	Failure consequence
Forest; domain; DC; OU; SID	ad.aster.example on AST-DC01	Wrong DNS or broken trust prevents fresh authentication and policy processing

Foundation concept: DNS

Problem solved: Resolve server names and locate AD services. Everyday analogy: A directory with addresses and service listings.

A/AAAA records locate hosts; SRV records locate services. AD-integrated DNS data is stored in directory partitions. DNS resolution is a dependency, not proof that a service port is open.

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Zone; A; PTR; SRV; FQDN	FS01 A record .20 and LDAP DC-locator records	Joining the domain fails even when ping by IP works

Foundation concept: DHCP

Problem solved: Assign consistent client network settings. Everyday analogy: A reception desk allocating temporary desks.

Discover, Offer, Request and Acknowledge establish a lease. Options carry DNS/domain settings. Infrastructure addresses remain outside the client pool.

Terms	Example in this build	Failure consequence
Scope; lease; exclusion; option 006; authorisation	192.168.50.100-149; DNS .10	APIPA or wrong DNS causes connectivity and domain failures

Foundation concept: SMB and NTFS

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Terms	Example in this build	Failure consequence
UNC; share ACL; NTFS ACL; inheritance; AGDLP	HR\$ and DL_FS_HR_Modify	Wrong grants cause either data exposure or legitimate access denial

Foundation concept: Group Policy

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Terms	Example in this build	Failure consequence
GPO; link; user scope; computer scope; resultant policy	HR Control Panel restriction	A created but unlinked policy has no effect

Foundation concept: Windows Server Backup

Problem solved: Recover a prior file state. Everyday analogy: A dated archive with a retrieval procedure.

WSB creates a recoverable backup using Windows backup/VSS facilities. Restore the selected version to another location and compare content. Reusing a remote destination may overwrite the previous backup.

Terms	Example in this build	Failure consequence
Recovery point; RPO; RTO; VSS; restore	FS01 data to BK01 restricted share	A job success message alone cannot prove restore integrity

Foundation concept: Firewall and event logs

Problem solved: Control connectivity and explain failures. Everyday analogy: A gatekeeper and incident diary.

Rules match profiles, directions, protocols and addresses. Logs record component events; timestamps and providers help connect a symptom to a cause.

Terms	Example in this build	Failure consequence
Profile; inbound; port; provider; event ID	SMB 445; Security/System/Application logs	Disabling all firewall profiles hides the fault and changes the security boundary

PowerShell starter and command-reading rules

Action	Exact method and reason
Open an elevated server console	Start > type Windows PowerShell > right-click Windows PowerShell > Run as administrator > Yes. On an English reference build the title begins Administrator.
Confirm the current machine	Run hostname, whoami and Get-Date. Match the task Perform on field before entering any command that changes settings.
Interpret symbols	A dash introduces a parameter; quoted text is a literal value; \$ introduces a variable; the pipe passes objects; # begins a comment. Read the line before running it.
Copy a long line	Use this HTML edition or the task command reference. PDF may wrap a line for printing; a display wrap is not an instruction to press Enter.
Private values	Read-Host -AsSecureString prompts for secrets without placing the password in the script. A regular Read-Host prompt is visible; do not capture its secret answer.
Choose one configuration route	Use the numbered GUI route or its command equivalent, then run verification. Running both creation routes can produce duplicate-object errors.
Read error output	Record host, time and full error; check the task Common errors table. Stop dependent tasks until the stated verification passes.
Commands in supplied files	Task reference files deliberately stop if run wholesale. They contain different host blocks and sometimes recovery actions. Open them in Notepad, read the host comments and copy only the block required now.

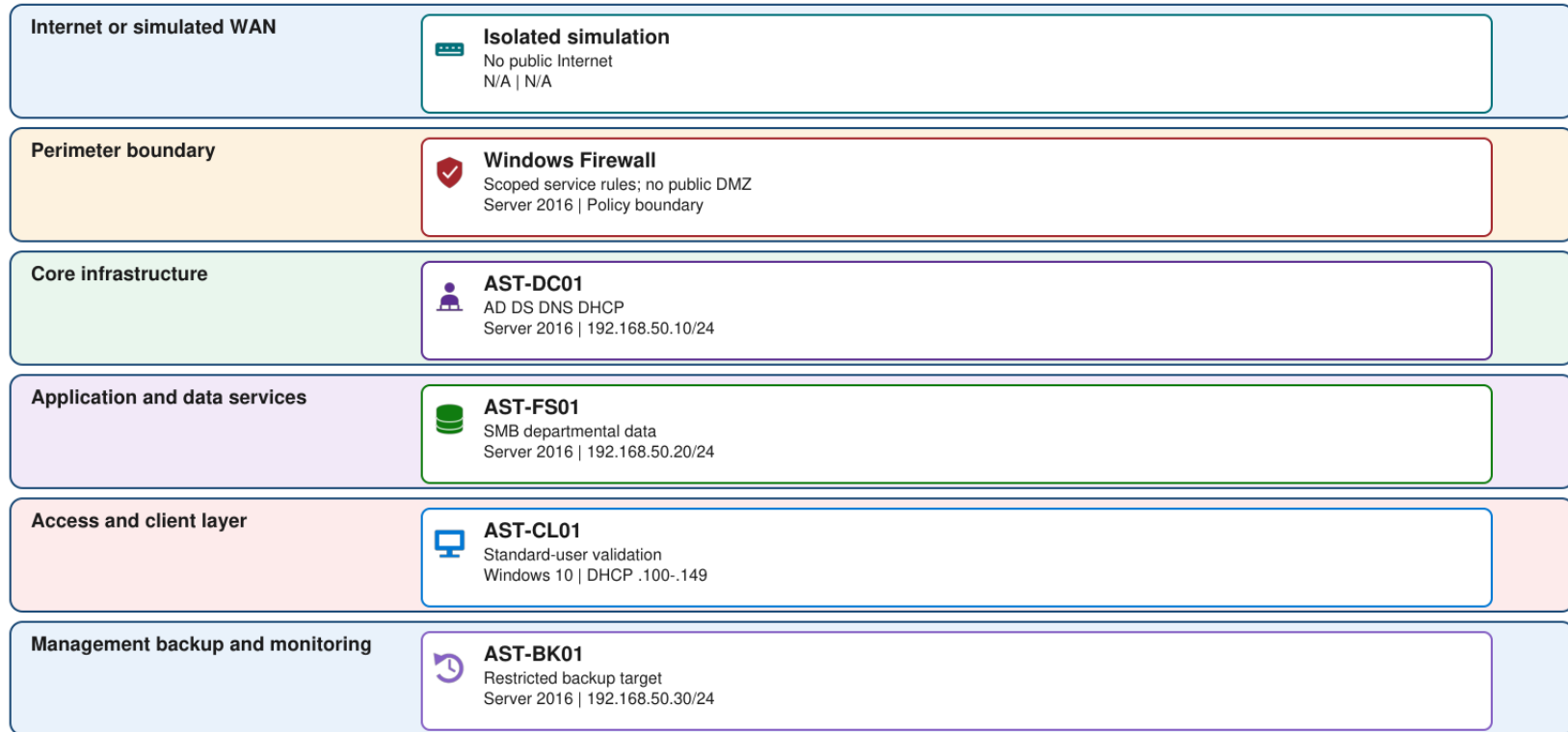
SECTION 3 ARCHITECTURE AND DIAGRAMS

The overview below explains the final service design. Detailed diagrams reappear immediately before the phase where they are used. Read the guide beneath each diagram; no separate diagram file is required for execution. Original service icons are embedded. Editable SVG, draw.io, XML and Mermaid sources remain in the optional assets.

D2 | Target service design

MCSA-P01 | D2 | Logical architecture in service layers

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Private lab addresses | Open the SVG at full size for readable labels

Legend solid: traffic dashed: replication
dotted: management red: recovery N/A: logical

Logical architecture in service layers

Current phase: Target service design. Read the horizontal bands as service responsibilities, not as separate VLANs. The foundation uses one LAN; Windows Firewall provides host-level controls. Exact packet exchanges are shown in D4 and D6. The management band names the backup target only when backup is implemented in this project.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

Diagram	Meaning	Where used
D1	Business context and ownership	Charter / design / relevant Section 5 phase
D2	Logical architecture in service layers	Charter / design / relevant Section 5 phase
D3	Physical host and VM placement	Charter / design / relevant Section 5 phase
D4	Network topology and service paths	Charter / design / relevant Section 5 phase
D5	Identity topology and policy scope	Charter / design / relevant Section 5 phase
D6	Client service flow from boot to file access	Charter / design / relevant Section 5 phase
D8	Backup and verified file restoration	Charter / design / relevant Section 5 phase
F3	Foundation: Physical host and VM placement	Charter / design / relevant Section 5 phase
F4	Foundation: Network topology and service paths	Charter / design / relevant Section 5 phase
F5	Foundation: Identity topology and policy scope	Charter / design / relevant Section 5 phase
F6	Foundation: Client service flow from boot to file access	Charter / design / relevant Section 5 phase

SECTION 4 LAB BUILD SHEET AND BILL OF MATERIALS

Extract the ZIP completely before opening any HTML guide. PDF is the print copy; HTML is the navigation and command-copy version; Markdown is editable source. The same content is included in each format. Diagrams are embedded here and beside the relevant implementation phase. PDF supports zoom; this HTML edition embeds every image and works as a single offline file. Editable diagram assets are optional.

This is an isolated legacy-platform training simulation, not a production deployment prescription. Use only institute-authorised or evaluation media; no operating-system ISO, product key or prebuilt VM is redistributed. Microsoft Server 2016 evaluation media is time limited. Windows 10 client media and its use rights must be supplied legally by the institute.

No Windows Server VM was executed in the authoring environment. Commands, diagrams and document consistency have been reviewed, but genuine screenshots and measured test results must be produced by the trainee. Items explicitly marked VERIFY IN YOUR BUILD describe UI or module differences that require inspection before changing settings.

Use fictional data only. Passwords, the DSRM secret, backupwriter password and VPN/DHCP shared secrets are chosen privately for this lab. Never put them in screenshots, scripts or the submission report. Keep recovery credentials in the institute-approved credential record.

Host requirement	Value
CPU	64-bit host with VT-x/AMD-V enabled; at least four logical cores preferred
RAM	16 GB reference host; full baseline 10 GB guest allocation
Storage	250 GB free SSD baseline; allow 300-350 GB for WDS/media and additional guests; monitor actual free space
Hypervisor	Legal VMware Workstation installation supported by host; nested virtualisation not needed
Network	VMnet2 isolated 192.168.50.0/24; no host adapter; VMware DHCP disabled
OS media	Server 2016 Desktop Experience evaluation/authorised media; Windows 10 domain-capable institute media

VM	Role	OS	CPU	RAM MB	Disks	Network	IP	GW	DNS
AST-DC01	AD DNS DHCP	Server 2016	2	2048	60 GB OS	VMnet2	192.168.50.10/24	None	.10
AST-FS01	File server	Server 2016	2	3072	60+20 GB	VMnet2	192.168.50.20/24	None	.10

VM	Role	OS	CPU	RAM MB	Disks	Network	IP	GW	DNS
AST-BK01	Backup target	Server 2016	2	2048	60+60 GB	VMnet2	192.168.50.30/24	None	.10
AST-CL01	Test client	Windows 10	2	3072	60 GB	VMnet2	DHCP .100-.149	None	.10

Active VM schedule: DC01 + FS01 + BK01 + CL01 can run together (10 GB guest RAM).

Use a separate project baseline or remove the previous specialist guest before reusing an IP. DEP01 and VPN01 intentionally both use .40 in different isolated project environments; they must not run together on the same LAN. If continuing all projects in one cumulative environment, plan and document new unique IPs before combining them.

Credential	Use and handling
Local Administrator / labadmin	OS installation and recovery before domain join; unique lab-only password
ASTER\Administrator	Lab infrastructure configuration only; standard-user tests must use departmental users
Department users	One unique password per account; first sign-in change completed before VPN tests
DSRM	Chosen during promotion and stored privately; not the domain administrator password
backupwriter	Local non-admin on BK01 for restricted backup share
VPN/DHCP secrets	Only relevant projects; privately entered; never included in evidence

Preflight checklist: ☐ virtualisation enabled; ☐ legal media available; ☐ required simultaneous RAM available; ☐ free disk space checked; ☐ isolated networks verified; ☐ no competing DHCP; ☐ host and guest clocks aligned; ☐ credential record secured; ☐ foundation acceptance passed; ☐ baseline evidence retained.

Service	Protocol and port	Where used
DNS	UDP and TCP 53	Domain and internal host resolution
DHCP	UDP 67 server and 68 client	Initial and renewed client leases
AD identity	Kerberos TCP/UDP 88; LDAP TCP/UDP 389; SMB TCP 445; RPC TCP135 plus dynamic TCP49152-65535; GC TCP3268 when used	Domain services; use role-installed firewall rules rather than an incomplete handmade AD rule set
Time	NTP UDP123	Domain time relationships

Service	Protocol and port	Where used
DHCP failover	TCP647	Project 4 partner communication
DFSR	RPC TCP135 and dynamic RPC ports	Project 2 member communication
IIS	HTTP TCP80	Project 3 isolated static portal
WDS	PXE UDP4011; TFTP initial UDP69 and negotiated data; DHCP UDP67/68; WDS RPC/SMB as used by deployment	Project 3: retain role-installed WDS firewall rules
VPN	IKE UDP500/4500; ESP IP protocol50; L2TP UDP1701 protected by IPsec	Project 5 isolated WAN
RDP	TCP3389 in this test	Project 5 internal target through VPN

SECTION 5 IMPLEMENTATION

This section contains the complete build from an empty lab. Projects 1-6 are independently assignable: Project 2 does not require completion of Project 1, and Project 3 does not require Project 2. Every project needs the foundation service state B01-B16, whose full steps are here. Recommended learning sequence: each trainee completes all six, then joins a capstone team. Six people may instead be assigned one specialist project each if each first builds or demonstrates the foundation; that does not give each person all six implementation experiences.

Fresh build: start B01 and execute every task. Verified existing baseline: demonstrate B16 and compare every Section 4 value, B11 accounts, B12 group nesting, B14 DHCP and B15 GPO. Record actual outputs; only then proceed to the specialist tasks. A failed check sends you to its numbered foundation task in this same document. Keep earlier project VMs powered off or in a separate virtual network to prevent duplicate names, IPs and DHCP servers.

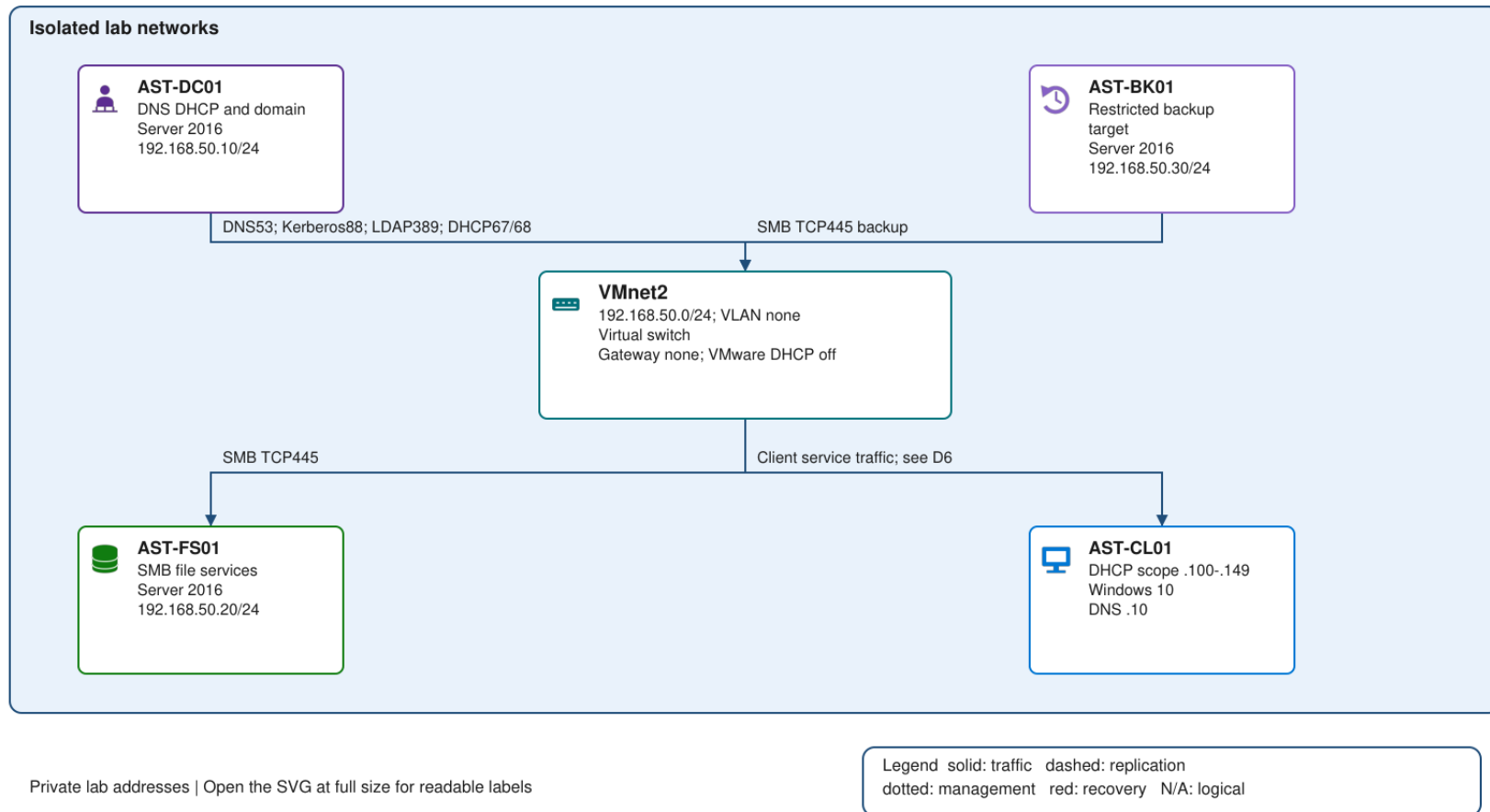
Task	Work	Depends on	Minutes
B01	Prepare the isolated virtual network	None	35
B02	Build AST-DC01	B01	60
B03	Build AST-FS01	B01	60
B04	Build AST-BK01	B01	60
B05	Build AST-CL01	B01	60
B06	Install Active Directory Domain Services	B02	20
B07	Create the forest and DNS service	B06	40
B08	Join AST-FS01 to the domain	B03 and B07	20
B09	Join AST-BK01 to the domain	B04 and B07	20
B10	Join AST-CL01 to the domain	B05 and B07	20
B11	Create the organisation and named accounts	B08 B09 B10	60
B12	Build group based access assignments	B11	40
B13	Publish three controlled departmental shares	B08 and B12	60

Task	Work	Depends on	Minutes
B14	Install and configure DHCP	B07	45
B15	Set domain account policy and HR user policy	B11 and B14	45
B16	Accept the foundation build	B13 B14 B15	60
P1-01	Prepare a restricted backup destination	B16	35
P1-02	Create test data and the first backup	P1-01	40
P1-03	Delete and restore a file with integrity proof	P1-02	40
P1-04	Prove retention and record the operational schedule	P1-03	30

F4 | Foundation phase A: isolated network

MCSA-P01 | F4 | Foundation: Network topology and service paths

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Foundation: Network topology and service paths

Current phase: Foundation phase A: isolated network. All endpoints attach to the same isolated subnet. Server IPs are outside the DHCP pool. There is no configured router or VLAN in the foundation. Port labels describe relevant flows; role-installed service rules must also allow dynamic RPC or TFTP exchanges where applicable.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

B01 Prepare the isolated virtual network

Field	Value
Objective and why	An isolated network prevents your lab DHCP server from answering real office clients.
Perform on	Physical host
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	35 minutes; add practice and evidence time
Depends on	None
Evidence produced	EVD-B01 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Sign in to the Windows host with a local administrator account. Open Task Manager > Performance > CPU.	Virtualization: Enabled	Hardware virtualisation is available. If Disabled, enable Intel VT-x or AMD-V in the host firmware before proceeding.	N
2	Obtain Windows Server 2016 evaluation ISO from Microsoft Evaluation Center and a legally supplied Windows 10 Pro or Enterprise ISO from your institute. Save both under a folder named C:\MCSA-Media.	Server: Desktop Experience; Client: domain-capable edition	Both ISO files exist. Installation media and licence keys are not included in this pack.	N
3	Install a legally obtained VMware Workstation version supported on your host. Open VMware Workstation > Edit > Virtual Network Editor > Change Settings.	Administrator approval on your own host	The editable network list appears. VERIFY IN YOUR BUILD: Workstation labels differ between versions.	N
4	Click Add Network. Select VMnet2 and click OK. Select Host-only. Clear Connect a host virtual adapter to this network and clear Use local DHCP service to distribute IP address to VMs.	Subnet IP 192.168.50.0; mask 255.255.255.0	VMnet2 is isolated from the host and has no VMware DHCP service.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Click Apply, then OK. Do not attach the training VMs to Bridged or NAT.	Every baseline VM: Custom > VMnet2	All guest traffic stays on the lab network.	N
6	Create a host folder C:\MCSA-Labs\Aster and a separate evidence folder C:\MCSA-Evidence. Open this ZIP after extracting it completely.	Do not open HTML directly inside the compressed ZIP	Relative links, diagrams and supplied files work.	Y: EVD-B01

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Host-side inventory only. VMware virtual network editing remains GUI-driven.
Get-CimInstance Win32_ComputerSystem | Select-Object TotalPhysicalMemory
Get-Volume -DriveLetter C
# There is no Windows Server cmdlet equivalent for VMware Virtual Network Editor.
```

Line	Command or value	Meaning and parameters
2	Get-CimInstance	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.

B Verification and interpretation

Inspect Virtual Network Editor and each VM Network Adapter setting.

Healthy example or criterion	Failed example or criterion
VMnet2 host-only; VMware DHCP disabled; 16 GB host RAM and adequate free SSD space.	VMnet8/NAT, Bridged, DHCP enabled or insufficient free space.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
VMnet2 cannot be edited	Editor is not elevated	Click Change Settings and approve your host administrator prompt.
No 64-bit guest choice	Hardware virtualisation disabled	Enable VT-x/AMD-V in firmware; restart the host.
Unexpected DHCP lease	A VM is on NAT/Bridged or VMware DHCP is enabled	Power off that VM; set Custom VMnet2; disable VMware DHCP on VMnet2.

D Rollback

No guest data exists yet. Remove only the newly created VMnet2 if unused. Do not remove networks belonging to other projects.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

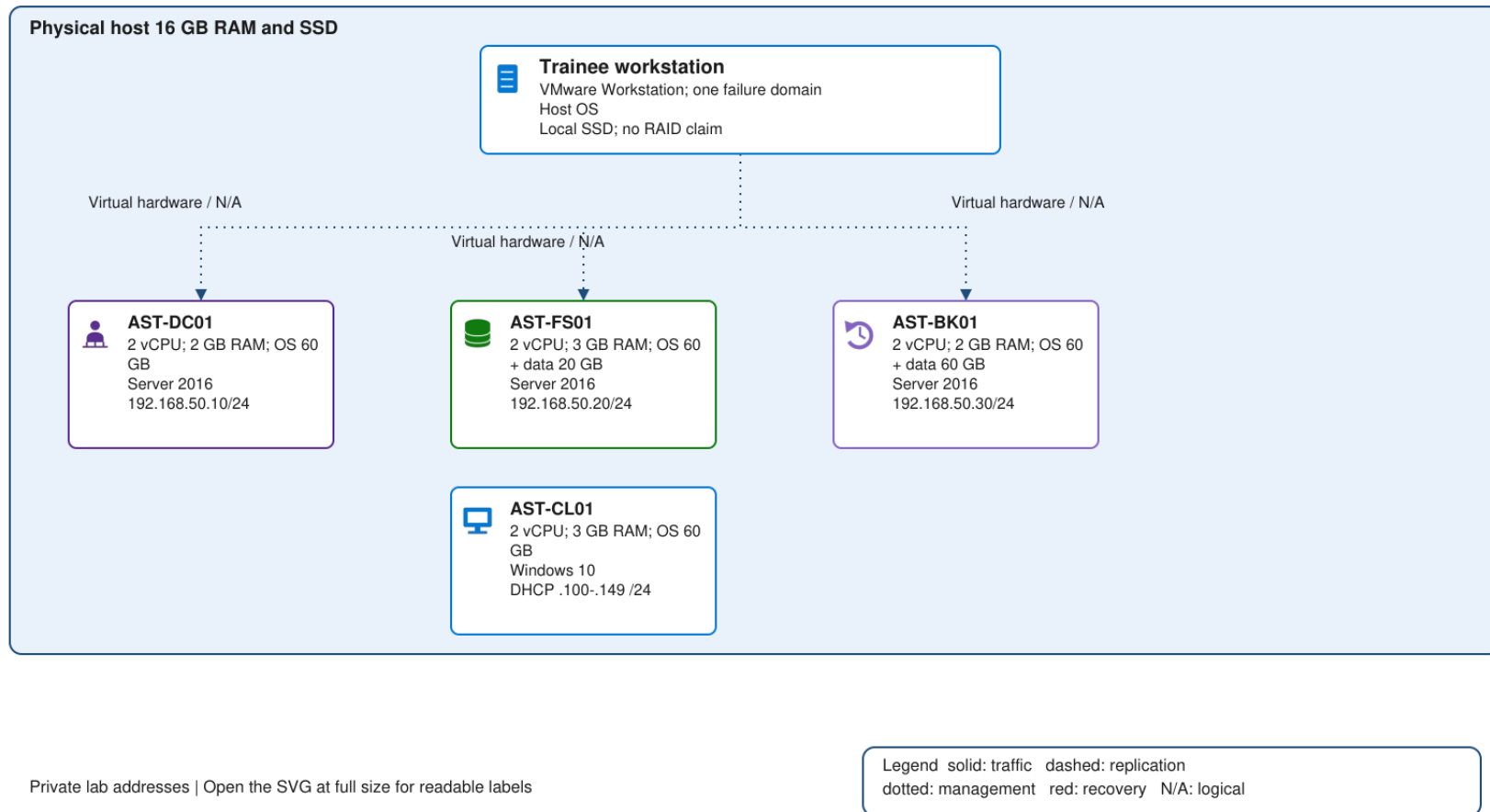
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

PHASE EXIT GATE: run the verification for every task in this phase; record actual output and evidence IDs in the tracker. Continue only when all are correct. All lab NICs must use the documented isolated network.

F3 | Foundation phase B: virtual machines

MCSA-P01 | F3 | Foundation: Physical host and VM placement

Version 1.0 | 16 September 2026 | Prepared by _____



Foundation: Physical host and VM placement

Current phase: Foundation phase B: virtual machines. All guests share the physical host, so a second VM does not remove the host failure domain. Disk figures are virtual capacities; snapshots and imported images consume additional physical space. Follow the active-VM schedule rather than powering on every guest in the collection.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

B02 Build AST-DC01

Field	Value
Objective and why	Create a predictable guest with a unique identity before installing services.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B01
Evidence produced	EVD-B02 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	VMware Workstation > File > New Virtual Machine > Typical (recommended) > Next. Choose I will install the operating system later > Next.	Avoid Easy Install	Guest operating system selection appears.	N
2	Choose Microsoft Windows and the matching version; click Next. Enter the VM name and location; click Next.	AST-DC01; C:\MCSA-Labs\Aster\AST-DC01; Windows Server 2016 Standard Evaluation (Desktop Experience)	Disk capacity page appears.	N
3	Set Maximum disk size to 60 GB. Select Store virtual disk as a single file; click Next. Click Customize Hardware.	Thin allocated virtual disk; 60 GB maximum	VM hardware list appears.	N
4	Select Memory, Processors, Network Adapter and New CD/DVD in turn. Set each exact value.	2048 MB RAM; 1 processor with 2 cores; Custom VMnet2; Use ISO image file and select the OS ISO	The summary shows RAM, two vCPUs, isolated NIC and selected ISO.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Click Close, then Finish. Power on this virtual machine. Press a key if asked to boot from CD/DVD.	Boot installation media	Windows Setup language page appears.	N
6	Select English language, your keyboard layout and time/currency preference. Click Next, then Install now.	English UI used throughout this guide	Edition selection or product-key page appears.	N
7	Choose the licensed/evaluation edition. For Server explicitly choose Desktop Experience. For client, use a legally supplied Pro/Enterprise licence or evaluation workflow.	Windows Server 2016 Standard Evaluation (Desktop Experience)	Licence terms appear. Windows Home cannot join this domain.	N
8	Accept the licence terms > Next. Choose Custom: Install Windows only (advanced). Select the empty 60 GB drive > Next.	Only the empty disk belonging to this new VM	Windows copies files and restarts. Do not start a second installation on the restart.	N
9	For Server, set the local Administrator password. For client, finish offline setup with local user labadmin; use an institute-provided lab-only password.	A unique lab-only password of at least 12 characters; record securely outside screenshots	Desktop appears. Never reuse a work or personal password.	N
10	Open Start > Windows System > Run; type sysdm.cpl > OK. On Computer Name click Change. Enter the exact name > OK > OK > Close > Restart Now.	AST-DC01	The guest restarts with its assigned computer name.	N
11	After sign-in open Run > ncpa.cpl. Right-click Ethernet > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties.	Use the following IP address	IPv4 fields are editable.	N
12	Enter IP, mask and DNS. Leave the default gateway blank. Click OK and Close.	192.168.50.10; 255.255.255.0; Preferred DNS 192.168.50.10; no alternate DNS	No real-world gateway is introduced. DC01 DNS will become available in B07.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
13	Open an elevated Windows PowerShell: right-click Start > Windows PowerShell (Admin), or Start > Windows PowerShell > right-click > Run as administrator.	Administrator console	The title says Administrator. Run the verification below.	Y: EVD-B02

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Run inside AST-DC01; use the actual adapter alias shown by Get-NetAdapter.
Get-NetAdapter
Rename-Computer -NewName 'AST-DC01' -Restart
# After restart, elevated PowerShell:
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress '192.168.50.10' -PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses '192.168.50.10'
# New-NetIPAddress is an alternative to GUI entry, not a command to rerun after GUI configuration.
```

Line	Command or value	Meaning and parameters
2	Get-NetAdapter	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.
3	Rename-Computer	Runs the task-specific command with the explicit values shown. Use Get-Help Rename-Computer -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
5	New-NetIPAddress	Assigns the exact static IPv4 address to the selected adapter; PrefixLength 24 means mask 255.255.255.0. Do not rerun after assigning it through the GUI.
6	Set-DnsClientServerAddress	Selects DNS resolvers for the named adapter; ResetServerAddresses restores DHCP-supplied resolvers.

B Verification and interpretation

```
hostname
ipconfig /all
Get-Volume
```

Healthy example or criterion	Failed example or criterion
AST-DC01; IPv4 192.168.50.10; DNS 192.168.50.10; NTFS data volume when specified.	Wrong hostname, APIPA 169.254 address, public DNS or a missing data volume.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Black console/no installer	ISO not connected or boot order wrong	Power off; VM Settings > CD/DVD > select correct ISO and Connect at power on; restart and press a key.
Server has only a command prompt	Server Core edition was installed	Recreate this empty lab VM using Desktop Experience; do not try to add a desktop to Server Core.
D: unavailable	Optical drive already owns D:	Use Disk Management to move only the DVD letter to Z:, then assign D: to the data disk.

D Rollback

Before domain joining, an incorrectly built empty VM can be shut down and recreated. Never delete a VM containing the only copy of evidence or data.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B03 Build AST-FS01

Field	Value
Objective and why	Create a predictable guest with a unique identity before installing services.
Perform on	AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B01
Evidence produced	EVD-B03 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	VMware Workstation > File > New Virtual Machine > Typical (recommended) > Next. Choose I will install the operating system later > Next.	Avoid Easy Install	Guest operating system selection appears.	N
2	Choose Microsoft Windows and the matching version; click Next. Enter the VM name and location; click Next.	AST-FS01; C:\MCSA-Labs\Aster\AST-FS01; Windows Server 2016 Standard Evaluation (Desktop Experience)	Disk capacity page appears.	N
3	Set Maximum disk size to 60 GB. Select Store virtual disk as a single file; click Next. Click Customize Hardware.	Thin allocated virtual disk; 60 GB maximum	VM hardware list appears.	N
4	Select Memory, Processors, Network Adapter and New CD/DVD in turn. Set each exact value.	3072 MB RAM; 1 processor with 2 cores; Custom VMnet2; Use ISO image file and select the OS ISO	The summary shows RAM, two vCPUs, isolated NIC and selected ISO.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Click Close, then Finish. Power on this virtual machine. Press a key if asked to boot from CD/DVD.	Boot installation media	Windows Setup language page appears.	N
6	Select English language, your keyboard layout and time/currency preference. Click Next, then Install now.	English UI used throughout this guide	Edition selection or product-key page appears.	N
7	Choose the licensed/evaluation edition. For Server explicitly choose Desktop Experience. For client, use a legally supplied Pro/Enterprise licence or evaluation workflow.	Windows Server 2016 Standard Evaluation (Desktop Experience)	Licence terms appear. Windows Home cannot join this domain.	N
8	Accept the licence terms > Next. Choose Custom: Install Windows only (advanced). Select the empty 60 GB drive > Next.	Only the empty disk belonging to this new VM	Windows copies files and restarts. Do not start a second installation on the restart.	N
9	For Server, set the local Administrator password. For client, finish offline setup with local user labadmin; use an institute-provided lab-only password.	A unique lab-only password of at least 12 characters; record securely outside screenshots	Desktop appears. Never reuse a work or personal password.	N
10	Open Start > Windows System > Run; type sysdm.cpl > OK. On Computer Name click Change. Enter the exact name > OK > OK > Close > Restart Now.	AST-FS01	The guest restarts with its assigned computer name.	N
11	After sign-in open Run > ncpa.cpl. Right-click Ethernet > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties.	Use the following IP address	IPv4 fields are editable.	N
12	Enter IP, mask and DNS. Leave the default gateway blank. Click OK and Close.	192.168.50.20; 255.255.255.0; Preferred DNS 192.168.50.10; no alternate DNS	No real-world gateway is introduced. DC01 DNS will become available in B07.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
13	Open an elevated Windows PowerShell: right-click Start > Windows PowerShell (Admin), or Start > Windows PowerShell > right-click > Run as administrator.	Administrator console	The title says Administrator. Run the verification below.	N
14	Shut down this guest. VMware > Edit virtual machine settings > Add > Hard Disk > Next > SCSI > Next > Create a new virtual disk > Next.	20 GB, single file; Finish; OK; power on	A second empty disk is attached.	N
15	Run diskmgmt.msc. If prompted to initialise Disk 1, select GPT > OK. Right-click its Unallocated space > New Simple Volume > Next.	20 GB empty disk only	The volume wizard opens. Check disk capacity before making changes.	N
16	Use the maximum volume size > Next. If the DVD drive already uses D:, cancel; right-click the DVD volume > Change Drive Letter and Paths > Change > choose Z: > OK. Restart New Simple Volume on the empty disk.	Assign data volume D:	The data disk receives D: and the optical drive no longer conflicts.	N
17	Choose NTFS, Allocation unit size Default, Volume label DATA, Perform a quick format. Click Next > Finish.	D: labelled DATA	Explorer shows the empty data disk.	Y: EVD-B03

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Run inside AST-FS01; use the actual adapter alias shown by Get-NetAdapter.
Get-NetAdapter
Rename-Computer -NewName 'AST-FS01' -Restart
# After restart, elevated PowerShell:
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress '192.168.50.20' -PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses '192.168.50.10'
# New-NetIPAddress is an alternative to GUI entry, not a command to rerun after GUI configuration.
# Only for the verified empty secondary disk; never select the OS disk.
Get-Disk | Format-Table Number,FriendlyName,Size,PartitionStyle
```

```
# Disk 1 must be the empty secondary disk before executing:
Initialize-Disk -Number 1 -PartitionStyle GPT
New-Partition -DiskNumber 1 -UseMaximumSize -DriveLetter D | Format-Volume -FileSystem NTFS -NewFileSystemLabel DATA
-Confirm:$false
```

Line	Command or value	Meaning and parameters
2	Get-NetAdapter	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.
3	Rename-Computer	Runs the task-specific command with the explicit values shown. Use Get-Help Rename-Computer -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
5	New-NetIPAddress	Assigns the exact static IPv4 address to the selected adapter; PrefixLength 24 means mask 255.255.255.0. Do not rerun after assigning it through the GUI.
6	Set-DnsClientServerAddress	Selects DNS resolvers for the named adapter; ResetServerAddresses restores DHCP-supplied resolvers.
11	Initialize-Disk	Initialises only the verified empty secondary disk. A wrong disk number can destroy unrelated data.
12	New-Partition	Creates a data partition on that verified disk and formats it NTFS; inspect disk number and capacity first.

B Verification and interpretation

```
hostname
ipconfig /all
Get-Volume
```

Healthy example or criterion	Failed example or criterion
AST-FS01; IPv4 192.168.50.20; DNS 192.168.50.10; NTFS data volume when specified.	Wrong hostname, APIPA 169.254 address, public DNS or a missing data volume.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Black console/no installer	ISO not connected or boot order wrong	Power off; VM Settings > CD/DVD > select correct ISO and Connect at power on; restart and press a key.
Server has only a command prompt	Server Core edition was installed	Recreate this empty lab VM using Desktop Experience; do not try to add a desktop to Server Core.
D: unavailable	Optical drive already owns D:	Use Disk Management to move only the DVD letter to Z:, then assign D: to the data disk.

D Rollback

Before domain joining, an incorrectly built empty VM can be shut down and recreated. Never delete a VM containing the only copy of evidence or data.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B04 Build AST-BK01

Field	Value
Objective and why	Create a predictable guest with a unique identity before installing services.
Perform on	AST-BK01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B01
Evidence produced	EVD-B04 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	VMware Workstation > File > New Virtual Machine > Typical (recommended) > Next. Choose I will install the operating system later > Next.	Avoid Easy Install	Guest operating system selection appears.	N
2	Choose Microsoft Windows and the matching version; click Next. Enter the VM name and location; click Next.	AST-BK01; C:\MCSA-Labs\Aster\AST-BK01; Windows Server 2016 Standard Evaluation (Desktop Experience)	Disk capacity page appears.	N
3	Set Maximum disk size to 60 GB. Select Store virtual disk as a single file; click Next. Click Customize Hardware.	Thin allocated virtual disk; 60 GB maximum	VM hardware list appears.	N
4	Select Memory, Processors, Network Adapter and New CD/DVD in turn. Set each exact value.	2048 MB RAM; 1 processor with 2 cores; Custom VMnet2; Use ISO image file and select the OS ISO	The summary shows RAM, two vCPUs, isolated NIC and selected ISO.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Click Close, then Finish. Power on this virtual machine. Press a key if asked to boot from CD/DVD.	Boot installation media	Windows Setup language page appears.	N
6	Select English language, your keyboard layout and time/currency preference. Click Next, then Install now.	English UI used throughout this guide	Edition selection or product-key page appears.	N
7	Choose the licensed/evaluation edition. For Server explicitly choose Desktop Experience. For client, use a legally supplied Pro/Enterprise licence or evaluation workflow.	Windows Server 2016 Standard Evaluation (Desktop Experience)	Licence terms appear. Windows Home cannot join this domain.	N
8	Accept the licence terms > Next. Choose Custom: Install Windows only (advanced). Select the empty 60 GB drive > Next.	Only the empty disk belonging to this new VM	Windows copies files and restarts. Do not start a second installation on the restart.	N
9	For Server, set the local Administrator password. For client, finish offline setup with local user labadmin; use an institute-provided lab-only password.	A unique lab-only password of at least 12 characters; record securely outside screenshots	Desktop appears. Never reuse a work or personal password.	N
10	Open Start > Windows System > Run; type sysdm.cpl > OK. On Computer Name click Change. Enter the exact name > OK > OK > Close > Restart Now.	AST-BK01	The guest restarts with its assigned computer name.	N
11	After sign-in open Run > ncpa.cpl. Right-click Ethernet > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties.	Use the following IP address	IPv4 fields are editable.	N
12	Enter IP, mask and DNS. Leave the default gateway blank. Click OK and Close.	192.168.50.30; 255.255.255.0; Preferred DNS 192.168.50.10; no alternate DNS	No real-world gateway is introduced. DC01 DNS will become available in B07.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
13	Open an elevated Windows PowerShell: right-click Start > Windows PowerShell (Admin), or Start > Windows PowerShell > right-click > Run as administrator.	Administrator console	The title says Administrator. Run the verification below.	N
14	Shut down this guest. VMware > Edit virtual machine settings > Add > Hard Disk > Next > SCSI > Next > Create a new virtual disk > Next.	60 GB, single file; Finish; OK; power on	A second empty disk is attached.	N
15	Run diskmgmt.msc. If prompted to initialise Disk 1, select GPT > OK. Right-click its Unallocated space > New Simple Volume > Next.	60 GB empty disk only	The volume wizard opens. Check disk capacity before making changes.	N
16	Use the maximum volume size > Next. If the DVD drive already uses D:, cancel; right-click the DVD volume > Change Drive Letter and Paths > Change > choose Z: > OK. Restart New Simple Volume on the empty disk.	Assign data volume D:	The data disk receives D: and the optical drive no longer conflicts.	N
17	Choose NTFS, Allocation unit size Default, Volume label DATA, Perform a quick format. Click Next > Finish.	D: labelled DATA	Explorer shows the empty data disk.	Y: EVD-B04

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Run inside AST-BK01; use the actual adapter alias shown by Get-NetAdapter.
Get-NetAdapter
Rename-Computer -NewName 'AST-BK01' -Restart
# After restart, elevated PowerShell:
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress '192.168.50.30' -PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses '192.168.50.10'
# New-NetIPAddress is an alternative to GUI entry, not a command to rerun after GUI configuration.
# Only for the verified empty secondary disk; never select the OS disk.
Get-Disk | Format-Table Number,FriendlyName,Size,PartitionStyle
```

```
# Disk 1 must be the empty secondary disk before executing:
Initialize-Disk -Number 1 -PartitionStyle GPT
New-Partition -DiskNumber 1 -UseMaximumSize -DriveLetter D | Format-Volume -FileSystem NTFS -NewFileSystemLabel DATA
    -Confirm:$false
```

Line	Command or value	Meaning and parameters
2	Get-NetAdapter	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.
3	Rename-Computer	Runs the task-specific command with the explicit values shown. Use Get-Help Rename-Computer -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
5	New-NetIPAddress	Assigns the exact static IPv4 address to the selected adapter; PrefixLength 24 means mask 255.255.255.0. Do not rerun after assigning it through the GUI.
6	Set-DnsClientServerAddress	Selects DNS resolvers for the named adapter; ResetServerAddresses restores DHCP-supplied resolvers.
11	Initialize-Disk	Initialises only the verified empty secondary disk. A wrong disk number can destroy unrelated data.
12	New-Partition	Creates a data partition on that verified disk and formats it NTFS; inspect disk number and capacity first.

B Verification and interpretation

```
hostname
ipconfig /all
Get-Volume
```

Healthy example or criterion	Failed example or criterion
AST-BK01; IPv4 192.168.50.30; DNS 192.168.50.10; NTFS data volume when specified.	Wrong hostname, APIPA 169.254 address, public DNS or a missing data volume.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Black console/no installer	ISO not connected or boot order wrong	Power off; VM Settings > CD/DVD > select correct ISO and Connect at power on; restart and press a key.
Server has only a command prompt	Server Core edition was installed	Recreate this empty lab VM using Desktop Experience; do not try to add a desktop to Server Core.
D: unavailable	Optical drive already owns D:	Use Disk Management to move only the DVD letter to Z:, then assign D: to the data disk.

D Rollback

Before domain joining, an incorrectly built empty VM can be shut down and recreated. Never delete a VM containing the only copy of evidence or data.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B05 Build AST-CL01

Field	Value
Objective and why	Create a predictable guest with a unique identity before installing services.
Perform on	AST-CL01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B01
Evidence produced	EVD-B05 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	VMware Workstation > File > New Virtual Machine > Typical (recommended) > Next. Choose I will install the operating system later > Next.	Avoid Easy Install	Guest operating system selection appears.	N
2	Choose Microsoft Windows and the matching version; click Next. Enter the VM name and location; click Next.	AST-CL01; C:\MCSA-Labs\Aster\AST-CL01; Windows 10 Pro or Enterprise	Disk capacity page appears.	N
3	Set Maximum disk size to 60 GB. Select Store virtual disk as a single file; click Next. Click Customize Hardware.	Thin allocated virtual disk; 60 GB maximum	VM hardware list appears.	N
4	Select Memory, Processors, Network Adapter and New CD/DVD in turn. Set each exact value.	3072 MB RAM; 1 processor with 2 cores; Custom VMnet2; Use ISO image file and select the OS ISO	The summary shows RAM, two vCPUs, isolated NIC and selected ISO.	N
5	Click Close, then Finish. Power on this virtual machine. Press a key if asked to boot from CD/DVD.	Boot installation media	Windows Setup language page appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	Select English language, your keyboard layout and time/currency preference. Click Next, then Install now.	English UI used throughout this guide	Edition selection or product-key page appears.	N
7	Choose the licensed/evaluation edition. For Server explicitly choose Desktop Experience. For client, use a legally supplied Pro/Enterprise licence or evaluation workflow.	Windows 10 Pro or Enterprise	Licence terms appear. Windows Home cannot join this domain.	N
8	Accept the licence terms > Next. Choose Custom: Install Windows only (advanced). Select the empty 60 GB drive > Next.	Only the empty disk belonging to this new VM	Windows copies files and restarts. Do not start a second installation on the restart.	N
9	For Server, set the local Administrator password. For client, finish offline setup with local user labadmin; use an institute-provided lab-only password.	A unique lab-only password of at least 12 characters; record securely outside screenshots	Desktop appears. Never reuse a work or personal password.	N
10	Open Start > Windows System > Run; type sysdm.cpl > OK. On Computer Name click Change. Enter the exact name > OK > OK > Close > Restart Now.	AST-CL01	The guest restarts with its assigned computer name.	N
11	After sign-in open Run > ncpa.cpl. Right-click Ethernet > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties.	Use the following IP address	IPv4 fields are editable.	N
12	Enter IP, mask and DNS. Leave the default gateway blank. Click OK and Close.	192.168.50.90; 255.255.255.0; Preferred DNS 192.168.50.10; no alternate DNS	No real-world gateway is introduced. DC01 DNS will become available in B07.	N
13	Open an elevated Windows PowerShell: right-click Start > Windows PowerShell (Admin), or Start > Windows PowerShell > right-click > Run as administrator.	Administrator console	The title says Administrator. Run the verification below.	Y: EVD-B05

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Run inside AST-CL01; use the actual adapter alias shown by Get-NetAdapter.
Get-NetAdapter
Rename-Computer -NewName 'AST-CL01' -Restart
# After restart, elevated PowerShell:
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress '192.168.50.90' -PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses '192.168.50.10'
# New-NetIPAddress is an alternative to GUI entry, not a command to rerun after GUI configuration.
```

Line	Command or value	Meaning and parameters
2	Get-NetAdapter	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.
3	Rename-Computer	Runs the task-specific command with the explicit values shown. Use Get-Help Rename-Computer -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
5	New-NetIPAddress	Assigns the exact static IPv4 address to the selected adapter; PrefixLength 24 means mask 255.255.255.0. Do not rerun after assigning it through the GUI.
6	Set-DnsClientServerAddress	Selects DNS resolvers for the named adapter; ResetServerAddresses restores DHCP-supplied resolvers.

B Verification and interpretation

```
hostname
ipconfig /all
Get-Volume
```

Healthy example or criterion	Failed example or criterion
AST-CL01; IPv4 192.168.50.90; DNS 192.168.50.10; NTFS data volume when specified.	Wrong hostname, APIPA 169.254 address, public DNS or a missing data volume.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Black console/no installer	ISO not connected or boot order wrong	Power off; VM Settings > CD/DVD > select correct ISO and Connect at power on; restart and press a key.
Server has only a command prompt	Server Core edition was installed	Recreate this empty lab VM using Desktop Experience; do not try to add a desktop to Server Core.
D: unavailable	Optical drive already owns D:	Use Disk Management to move only the DVD letter to Z:, then assign D: to the data disk.

D Rollback

Before domain joining, an incorrectly built empty VM can be shut down and recreated. Never delete a VM containing the only copy of evidence or data.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

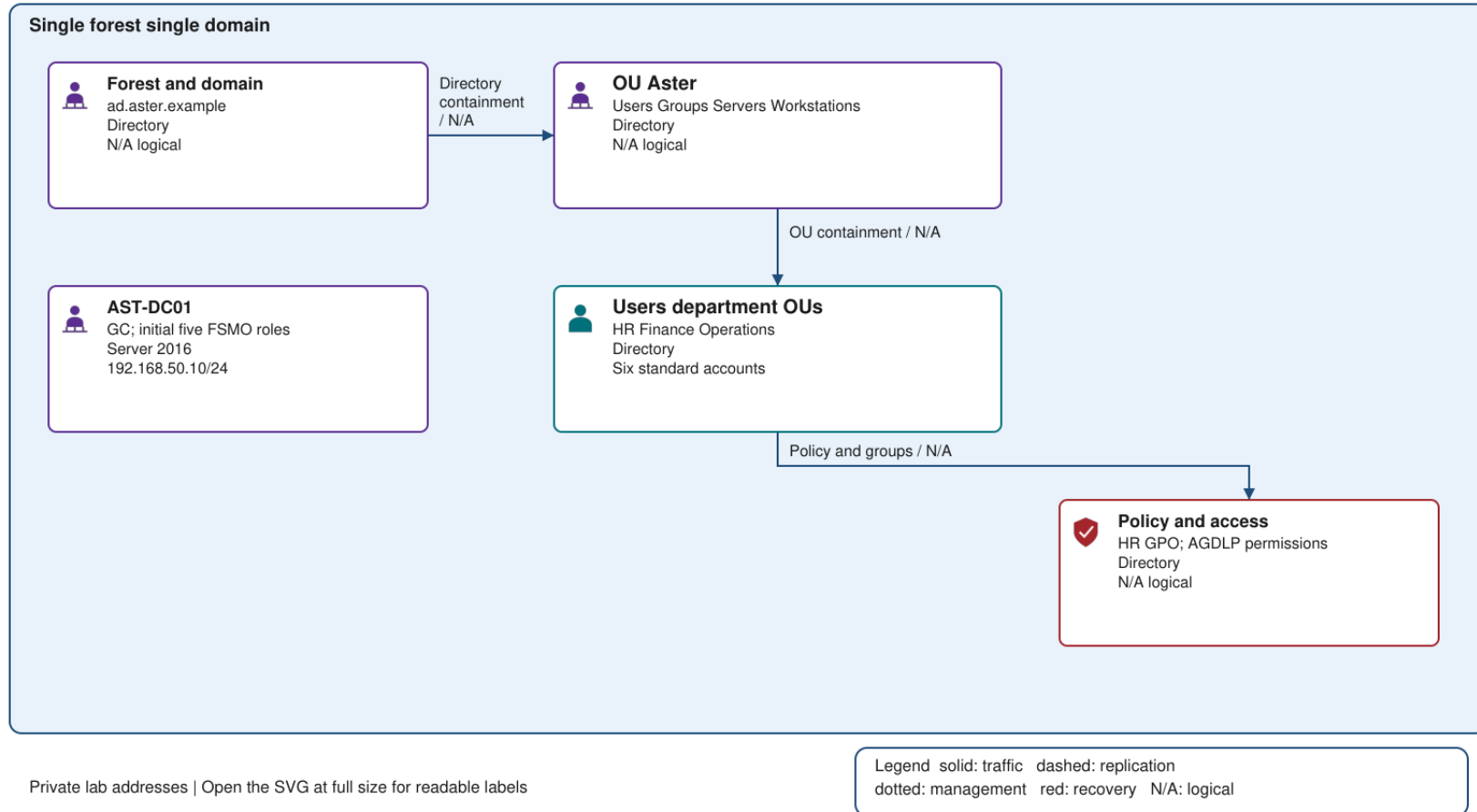
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

PHASE EXIT GATE: run the verification for every task in this phase; record actual output and evidence IDs in the tracker. Continue only when all are correct. All four consoles, hostnames, disks and static addresses must match the inventory.

F5 | Foundation phase C: directory and membership

MCSA-P01 | F5 | Foundation: Identity topology and policy scope

Version 1.0 | 16 September 2026 | Prepared by _____



Foundation: Identity topology and policy scope

Current phase: Foundation phase C: directory and membership. OU containment arrows are logical relationships and have no protocol port. DC objects remain in the built-in Domain Controllers OU. Groups grant file permissions; OUs organise policy scope. Only Project 4 adds a second DC and proves replication.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

B06 Install Active Directory Domain Services

Field	Value
Objective and why	The role binaries must exist before the server can become a domain controller.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	20 minutes; add practice and evidence time
Depends on	B02
Evidence produced	EVD-B06 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Open the VMware console for AST-DC01. Sign in with the task administrator account. Open PowerShell and run hostname. Close the console after checking the name.	AST-DC01	The computer name matches the target; no other VM is being changed.	N
2	Start > Server Manager > Manage > Add Roles and Features. On [Before You Begin] click Next.	Run on AST-DC01	[Select installation type] appears.	N
3	[Select installation type]: select Role-based or feature-based installation. Click Next.	Role-based or feature-based installation	[Select destination server] appears.	N
4	[Select destination server]: select Select a server from the server pool. Highlight the named local host. Click Next.	AST-DC01	[Select server roles] appears.	N
5	[Select server roles]: tick Active Directory Domain Services. If the required-features dialog opens, leave Include management tools selected and click Add Features.	Active Directory Domain Services	The role checkbox is selected.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	On [Select server roles] click Next. On [Select features], keep the automatically selected supporting features and click Next.	Do not select unrelated optional features	The role information page appears, or file services proceed directly to confirmation.	N
7	On [Active Directory Domain Services], read the role description and click Next.	No setting is changed on this information page	[Confirm installation selections] appears.	N
8	[Confirm installation selections]: confirm the target hostname and selected roles. Leave Restart the destination server automatically if required cleared. Click Install. On [Installation progress], wait for Installation succeeded on AST-DC01. Click Close.	Only AST-DC01; selected services: Active Directory Domain Services	The role and management console are installed. Configuration follows in the next numbered step or task.	Y: EVD-B06

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
```

Line	Command or value	Meaning and parameters
1	Install-WindowsFeature	Installs only the named Windows roles/features; IncludeManagementTools adds the corresponding management consoles.

B Verification and interpretation

```
Get-WindowsFeature AD-Domain-Services
```

Healthy example or criterion	Failed example or criterion
Install State: Installed. The server is not yet a domain controller.	Available means binaries are not installed.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
AD DS role installation is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

D Rollback

If promotion has not occurred, Server Manager > Manage > Remove Roles and Features can remove AD DS. After promotion, demote properly; do not just remove the role.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B07 Create the forest and DNS service

Field	Value
Objective and why	AD requires an initial domain, a directory database and DNS service records.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	40 minutes; add practice and evidence time
Depends on	B06
Evidence produced	EVD-B07 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Server Manager > notification flag > Promote this server to a domain controller. Deployment Configuration: select Add a new forest.	Root domain name: ad.aster.example; Next	Domain Controller Options appears.	N
2	Set forest and domain functional level to Windows Server 2016. Keep DNS server and Global Catalog checked. Type and confirm a unique Directory Services Restore Mode password.	Record DSRM password privately; Next	DNS Options appears.	N
3	On DNS Options read the delegation warning. This isolated new example forest has no parent DNS delegation. Click Next.	Do not create an external delegation	Additional Options appears.	N
4	Confirm NetBIOS domain name ASTER. Click Next. On Paths leave database, logs and SYSVOL at their defaults. Click Next.	C:\Windows\NTDS and C:\Windows\SYSVOL	Review Options appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Review Options: verify the domain name and click Next. Prerequisites Check: require successful checks; review expected delegation warning. Click Install.	ad.aster.example	Server promotes and automatically restarts.	N
6	Sign in as ASTER\Administrator. Run dsa.msc; expand the domain. Run dnsmgmt.msc; expand Forward Lookup Zones.	ad.aster.example and _msdcs.ad.aster.example	Directory and DNS consoles show the new forest.	Y: EVD-B07

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
$dsrm=Read-Host 'Unique lab DSRM password' -AsSecureString
Install-ADDSForest -DomainName 'ad.aster.example' -DomainNetbiosName 'ASTER' -ForestMode WinThreshold -DomainMode WinThreshold -InstallDns -SafeModeAdministratorPassword $dsrm -Force
# WinThreshold is the PowerShell enum for the Windows Server 2016 functional level.
```

Line	Command or value	Meaning and parameters
1	\$dsrm=Read-Host	Stores a named value or result used by a later line. Read the assigned path/value before execution; a prompt must use your actual observed result.
2	Install-ADDSForest	Creates the initial forest/domain with DNS and a prompted DSRM recovery password. It restarts the server.

B Verification and interpretation

```
Get-ADDomain | Select-Object DNSRoot,NetBIOSName
Get-ADForest | Select-Object RootDomain
net share
nslookup -type=SRV _ldap._tcp.dc._msdcs.ad.aster.example
```

Healthy example or criterion	Failed example or criterion
DNSRoot ad.aster.example; SYSVOL and NETLOGON shares; LDAP SRV answer identifies AST-DC01 on port 389.	NXDOMAIN, no SYSVOL/NETLOGON, or a different DNSRoot.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
DNS delegation warning	No parent zone exists in the isolated lab	Proceed only when the remaining prerequisites succeed and the domain is exactly ad.aster.example.
Promotion prerequisites fail	IP, DNS or password configuration invalid	Verify B02 static IP and DNS; read the specific prerequisite error and correct that setting.
Domain tools unavailable after restart	Signed in too early or promotion failed	Wait for AD services; inspect Directory Service and DFS Replication logs and the promotion result.

D Rollback

This is the only DC. Demotion deletes this lab domain. Preserve evidence first; if the initial forest was created with the wrong name, rebuild the empty baseline from B02 rather than modifying a populated domain.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B08 Join AST-FS01 to the domain

Field	Value
Objective and why	A computer account establishes the trust needed for domain sign-in and policy.
Perform on	AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	20 minutes; add practice and evidence time
Depends on	B03 and B07
Evidence produced	EVD-B08 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Sign in with the local administrator. Open elevated PowerShell and run the DNS precheck.	nslookup ad.aster.example 192.168.50.10	The domain resolves. Do not join while DNS fails.	N
2	Run sysdm.cpl > Computer Name > Change. Under Member of select Domain.	ad.aster.example; OK	Domain credentials prompt appears.	N
3	Enter the domain administrator credentials.	ASTER\Administrator and its lab password	Welcome to the ad.aster.example domain appears.	N
4	Click OK on the welcome and restart prompts; click Close > Restart Now.	Restart this VM only	Sign-in screen returns.	N
5	Choose Other user and sign in using the domain account. Open PowerShell.	ASTER\Administrator	whoami displays aster\administrator; the machine remains named AST-FS01.	Y: EVD-B08

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Add-Computer -DomainName 'ad.aster.example' -Credential (Get-Credential 'ASTER\Administrator') -Restart
```

Line	Command or value	Meaning and parameters
1	Add-Computer	Joins the named domain using prompted credentials; Restart restarts this guest to complete the join.

B Verification and interpretation

```
whoami
(Get-CimInstance Win32_ComputerSystem).Domain
nltest /dsgetdc:ad.aster.example
```

Healthy example or criterion	Failed example or criterion
Domain ad.aster.example; nltest returns a DC name and successful status.	Domain WORKGROUP; error 1355/no domain controller found.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Domain not found	Client DNS points elsewhere	ncpa.cpl > IPv4 Properties > preferred DNS 192.168.50.10; run ipconfig /flushdns and repeat precheck.
Access denied	Wrong credentials	Use ASTER\Administrator, not the local Administrator of this member.
Clock differs	Kerberos prerequisites unmet	Correct date/time on the host and guests; align the client with DC01, then retry.

D Rollback

Before unjoining, verify a working local administrator account. Use sysdm.cpl > Change > Workgroup LAB > supply domain administrator credentials > restart. Delete only this test computer account from AD if rebuilding.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B09 Join AST-BK01 to the domain

Field	Value
Objective and why	A computer account establishes the trust needed for domain sign-in and policy.
Perform on	AST-BK01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	20 minutes; add practice and evidence time
Depends on	B04 and B07
Evidence produced	EVD-B09 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Sign in with the local administrator. Open elevated PowerShell and run the DNS precheck.	nslookup ad.aster.example 192.168.50.10	The domain resolves. Do not join while DNS fails.	N
2	Run sysdm.cpl > Computer Name > Change. Under Member of select Domain.	ad.aster.example; OK	Domain credentials prompt appears.	N
3	Enter the domain administrator credentials.	ASTER\Administrator and its lab password	Welcome to the ad.aster.example domain appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
4	Click OK on the welcome and restart prompts; click Close > Restart Now.	Restart this VM only	Sign-in screen returns.	N
5	Choose Other user and sign in using the domain account. Open PowerShell.	ASTER\Administrator	whoami displays aster\administrator; the machine remains named AST-BK01.	Y: EVD-B09

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Add-Computer -DomainName 'ad.aster.example' -Credential (Get-Credential 'ASTER\Administrator') -Restart
```

Line	Command or value	Meaning and parameters
1	Add-Computer	Joins the named domain using prompted credentials; Restart restarts this guest to complete the join.

B Verification and interpretation

```
whoami
(Get-CimInstance Win32_ComputerSystem).Domain
nltest /dsgetdc:ad.aster.example
```

Healthy example or criterion	Failed example or criterion
Domain ad.aster.example; nltest returns a DC name and successful status.	Domain WORKGROUP; error 1355/no domain controller found.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Domain not found	Client DNS points elsewhere	ncpa.cpl > IPv4 Properties > preferred DNS 192.168.50.10; run ipconfig /flushdns and repeat precheck.
Access denied	Wrong credentials	Use ASTER\Administrator, not the local Administrator of this member.
Clock differs	Kerberos prerequisites unmet	Correct date/time on the host and guests; align the client with DC01, then retry.

D Rollback

Before unjoining, verify a working local administrator account. Use sysdm.cpl > Change > Workgroup LAB > supply domain administrator credentials > restart. Delete only this test computer account from AD if rebuilding.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B10 Join AST-CL01 to the domain

Field	Value
Objective and why	A computer account establishes the trust needed for domain sign-in and policy.
Perform on	AST-CL01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	20 minutes; add practice and evidence time
Depends on	B05 and B07
Evidence produced	EVD-B10 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Sign in with the local administrator. Open elevated PowerShell and run the DNS precheck.	nslookup ad.aster.example 192.168.50.10	The domain resolves. Do not join while DNS fails.	N
2	Run sysdm.cpl > Computer Name > Change. Under Member of select Domain.	ad.aster.example; OK	Domain credentials prompt appears.	N
3	Enter the domain administrator credentials.	ASTER\Administrator and its lab password	Welcome to the ad.aster.example domain appears.	N
4	Click OK on the welcome and restart prompts; click Close > Restart Now.	Restart this VM only	Sign-in screen returns.	N
5	Choose Other user and sign in using the domain account. Open PowerShell.	ASTER\Administrator	whoami displays aster\administrator; the machine remains named AST-CL01.	Y: EVD-B10

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Add-Computer -DomainName 'ad.aster.example' -Credential (Get-Credential 'ASTER\Administrator') -Restart
```

Line	Command or value	Meaning and parameters
1	Add-Computer	Joins the named domain using prompted credentials; Restart restarts this guest to complete the join.

B Verification and interpretation

```
whoami
(Get-CimInstance Win32_ComputerSystem).Domain
nltest /dsgetdc:ad.aster.example
```

Healthy example or criterion	Failed example or criterion
Domain ad.aster.example; nltest returns a DC name and successful status.	Domain WORKGROUP; error 1355/no domain controller found.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Domain not found	Client DNS points elsewhere	ncpa.cpl > IPv4 Properties > preferred DNS 192.168.50.10; run ipconfig /flushdns and repeat precheck.
Access denied	Wrong credentials	Use ASTER\Administrator, not the local Administrator of this member.
Clock differs	Kerberos prerequisites unmet	Correct date/time on the host and guests; align the client with DC01, then retry.

D Rollback

Before unjoining, verify a working local administrator account. Use sysdm.cpl > Change > Workgroup LAB > supply domain administrator credentials > restart. Delete only this test computer account from AD if rebuilding.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

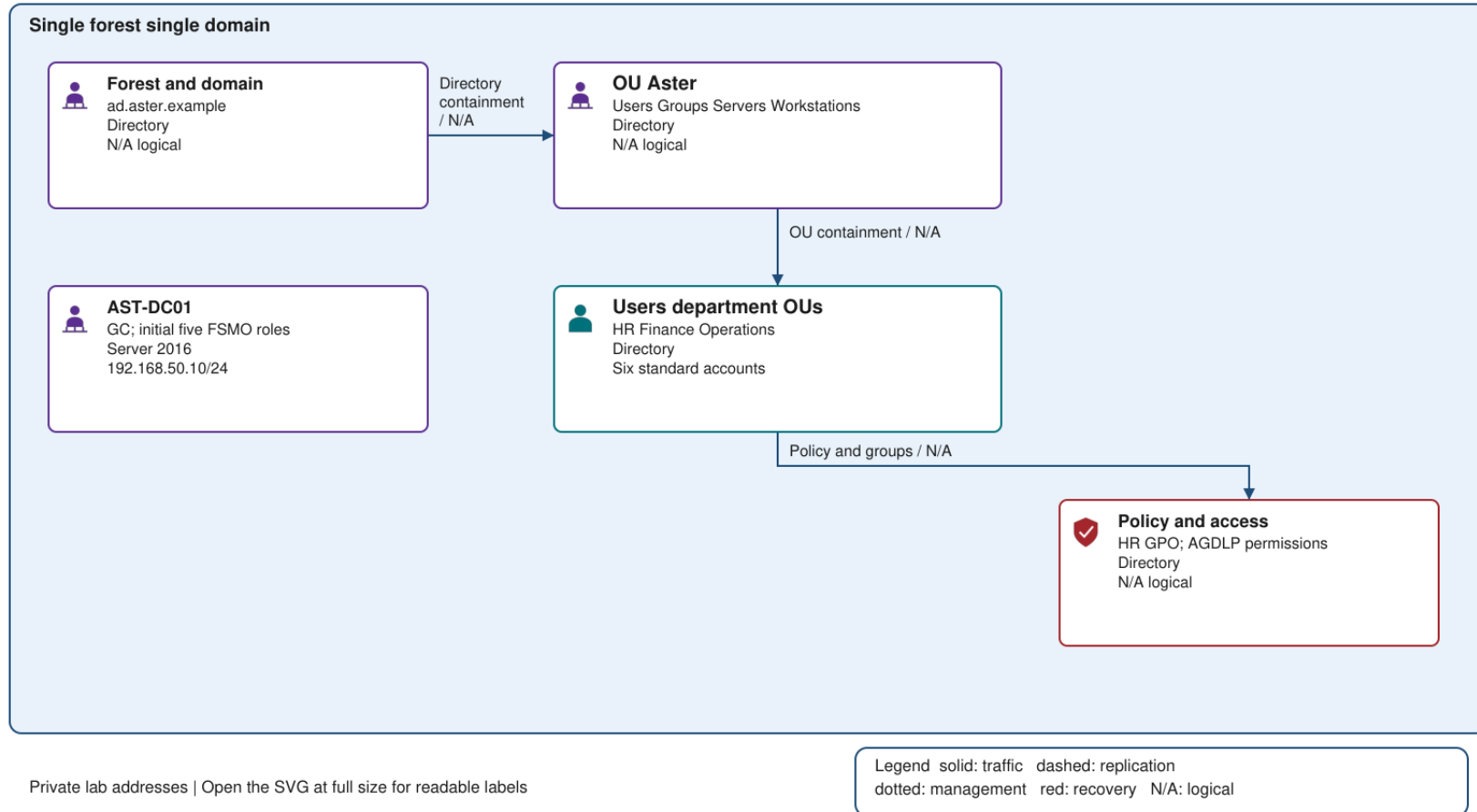
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

PHASE EXIT GATE: run the verification for every task in this phase; record actual output and evidence IDs in the tracker. Continue only when all are correct. Members and client must locate and join ad.aster.example through internal DNS.

F5 | Foundation phase D: users, groups and permissions

MCSA-P01 | F5 | Foundation: Identity topology and policy scope

Version 1.0 | 16 September 2026 | Prepared by _____



Foundation: Identity topology and policy scope

Current phase: Foundation phase D: users, groups and permissions. OU containment arrows are logical relationships and have no protocol port. DC objects remain in the built-in Domain Controllers OU. Groups grant file permissions; OUs organise policy scope. Only Project 4 adds a second DC and proves replication.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

B11 Create the organisation and named accounts

Field	Value
Objective and why	Separate user and computer locations make policy scope and ownership visible.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B08 B09 B10
Evidence produced	EVD-B11 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	On DC01 open Start > Windows Administrative Tools > Active Directory Users and Computers, or Run dsa.msc.	ASTER\Administrator	ad.aster.example is visible.	N
2	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Aster under DC=ad,DC=aster, DC=example	OU Aster exists under the specified parent.	N
3	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Users under OU=Aster,DC=ad, DC=aster,DC=example	OU Users exists under the specified parent.	N
4	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Groups under OU=Aster,DC=a d,DC=aster,DC=example	OU Groups exists under the specified parent.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Servers under OU=Aster,DC=ad,DC=aster,DC=example	OU Servers exists under the specified parent.	N
6	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Workstations under OU=Aster, DC=ad,DC=aster,DC=example	OU Workstations exists under the specified parent.	N
7	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	AdminAccounts under OU=Aster,DC=ad,DC=aster,DC=example	OU AdminAccounts exists under the specified parent.	N
8	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	HR under OU=Users,OU=Aster,DC=ad,DC=aster,DC=example	OU HR exists under the specified parent.	N
9	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Finance under OU=Users,OU=Aster,DC=ad,DC=aster,DC=example	OU Finance exists under the specified parent.	N
10	In ADUC right-click the container identified by this parent path > New > Organizational Unit. Leave Protect container from accidental deletion checked; click OK.	Operations under OU=Users,OU=Aster,DC=ad,DC=aster,DC=example	OU Operations exists under the specified parent.	N
11	In ADUC expand Aster > Users > HR. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Anita HR; anita.hr; suffix @ad.aster.example	User password page appears.	N
12	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	anita.hr; password privately recorded	Enabled user appears in HR OU.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
13	In ADUC expand Aster > Users > HR. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Vikas HR; vikas.hr; suffix @ad.aster.example	User password page appears.	N
14	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	vikas.hr; password privately recorded	Enabled user appears in HR OU.	N
15	In ADUC expand Aster > Users > Finance. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Rohan Finance; rohan.fin; suffix @ad.aster.example	User password page appears.	N
16	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	rohan.fin; password privately recorded	Enabled user appears in Finance OU.	N
17	In ADUC expand Aster > Users > Finance. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Neha Finance; neha.fin; suffix @ad.aster.example	User password page appears.	N
18	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	neha.fin; password privately recorded	Enabled user appears in Finance OU.	N
19	In ADUC expand Aster > Users > Operations. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Meera Operations; meera.ops; suffix @ad.aster.example	User password page appears.	N
20	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	meera.ops; password privately recorded	Enabled user appears in Operations OU.	N
21	In ADUC expand Aster > Users > Operations. Right-click the OU > New > User. Fill Full name and User logon name. Click Next.	Arjun Operations; arjun.ops; suffix @ad.aster.example	User password page appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
22	Enter and confirm a unique lab-only password. Keep User must change password at next logon checked; leave Account is disabled clear. Click Next > Finish.	arjun.ops; password privately recorded	Enabled user appears in Operations OU.	N
23	In ADUC open Computers. Right-click AST-FS01 > Move. Select Aster > Servers > OK.	AST-FS01 to Servers	The computer object is in its target OU. DC01 stays in Domain Controllers.	N
24	In ADUC open Computers. Right-click AST-BK01 > Move. Select Aster > Servers > OK.	AST-BK01 to Servers	The computer object is in its target OU. DC01 stays in Domain Controllers.	N
25	In ADUC open Computers. Right-click AST-CL01 > Move. Select Aster > Workstations > OK.	AST-CL01 to Workstations	The computer object is in its target OU. DC01 stays in Domain Controllers.	Y: EVD-B11

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Import-Module ActiveDirectory
New-ADOrganizationalUnit -Name 'Aster' -Path 'DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Users' -Path 'OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Groups' -Path 'OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Servers' -Path 'OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Workstations' -Path 'OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'AdminAccounts' -Path 'OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'HR' -Path 'OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Finance' -Path 'OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADOrganizationalUnit -Name 'Operations' -Path 'OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -ProtectedFromAccidentalDeletion $true
New-ADUser -Name 'Anita HR' -SamAccountName 'anita.hr' -UserPrincipalName 'anita.hr@ad.aster.example' -Path 'OU=HR,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for anita.hr')
```

```

-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true
New-ADUser -Name 'Vikas HR' -SamAccountName 'vikas.hr' -UserPrincipalName 'vikas.hr@ad.aster.example' -Path
'OU=HR,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for vikas.hr'
-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true

New-ADUser -Name 'Rohan Finance' -SamAccountName 'rohan.fin' -UserPrincipalName 'rohan.fin@ad.aster.example' -Path
'OU=Finance,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for rohan.fin'
-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true
New-ADUser -Name 'Neha Finance' -SamAccountName 'neha.fin' -UserPrincipalName 'neha.fin@ad.aster.example' -Path
'OU=Finance,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for neha.fin'
-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true
New-ADUser -Name 'Meera Operations' -SamAccountName 'meera.ops' -UserPrincipalName 'meera.ops@ad.aster.example' -Path
'OU=Operations,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for meera.ops'
-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true
New-ADUser -Name 'Arjun Operations' -SamAccountName 'arjun.ops' -UserPrincipalName 'arjun.ops@ad.aster.example' -Path
'OU=Operations,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' -AccountPassword (Read-Host 'Password for arjun.ops'
-AsSecureString) -Enabled $true -ChangePasswordAtLogon $true
Get-ADComputer 'AST-FS01' | Move-ADObject -TargetPath 'OU=Servers,OU=Aster,DC=ad,DC=aster,DC=example'
Get-ADComputer 'AST-BK01' | Move-ADObject -TargetPath 'OU=Servers,OU=Aster,DC=ad,DC=aster,DC=example'
Get-ADComputer 'AST-CL01' | Move-ADObject -TargetPath 'OU=Workstations,OU=Aster,DC=ad,DC=aster,DC=example'

```

Line	Command or value	Meaning and parameters
1	Import-Module	Runs the task-specific command with the explicit values shown. Use Get-Help Import-Module -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
2	New-ADOrganizationalUnit	Creates one OU under the exact distinguished-name parent; protection prevents accidental deletion.
11	New-ADUser	Creates the specified account, UPN and OU placement using a securely prompted initial password.
17	Get-ADComputer	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.

B Verification and interpretation

```

Get-ADUser -Filter * -SearchBase 'OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' | Select-Object SamAccountName,Enabled
Get-ADComputer -Filter * | Select-Object Name,DistinguishedName

```

Healthy example or criterion	Failed example or criterion
Six departmental accounts enabled; three member computer objects in the documented OUs.	Missing users, disabled accounts or members left in Computers.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
ADUC is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

D Rollback

Remove only unused test users after recording them. OUs protected from accidental deletion require View > Advanced Features > OU Properties > Object > clear protection before deleting an empty OU. Never delete Domain Controllers.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B12 Build group based access assignments

Field	Value
Objective and why	AGDLP separates people, roles and resource permissions, making access changes auditable.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	40 minutes; add practice and evidence time
Depends on	B11
Evidence produced	EVD-B12 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	GG_HR_Users; Global; Security	Group exists with the selected scope.	N
2	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	DL_FS_HR_Modify; Domain Local; Security	Group exists with the selected scope.	N
3	Double-click GG_HR_Users > Members > Add. Type the two accounts separated by a semicolon > Check Names > OK > Apply > OK.	anita.hr; vikas.hr	Both users appear in the global group.	N
4	Double-click DL_FS_HR_Modify > Members > Add. Type the global group > Check Names > OK > Apply > OK.	GG_HR_Users	The global group is nested in the resource group.	N
5	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	GG_Finance_Users; Global; Security	Group exists with the selected scope.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	DL_FS_Finance_Modify; Domain Local; Security	Group exists with the selected scope.	N
7	Double-click GG_Finance_Users > Members > Add. Type the two accounts separated by a semicolon > Check Names > OK > Apply > OK.	rohan.fin; neha.fin	Both users appear in the global group.	N
8	Double-click DL_FS_Finance_Modify > Members > Add. Type the global group > Check Names > OK > Apply > OK.	GG_Finance_Users	The global group is nested in the resource group.	N
9	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	GG_Operations_Users; Global; Security	Group exists with the selected scope.	N
10	DC01 > dsa.msc > Aster > Groups. Right-click > New > Group. Type the name, select the scope and Security; click OK.	DL_FS_Operations_Modify; Domain Local; Security	Group exists with the selected scope.	N
11	Double-click GG_Operations_Users > Members > Add. Type the two accounts separated by a semicolon > Check Names > OK > Apply > OK.	meera.ops; arjun.ops	Both users appear in the global group.	N
12	Double-click DL_FS_Operations_Modify > Members > Add. Type the global group > Check Names > OK > Apply > OK.	GG_Operations_Users	The global group is nested in the resource group.	Y: EVD-B12

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Import-Module ActiveDirectory
New-ADGroup -Name 'GG_HR_Users' -GroupScope Global -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
```

```

New-ADGroup -Name 'DL_FS_HR_Modify' -GroupScope DomainLocal -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
Add-ADGroupMember -Identity 'GG_HR_Users' -Members 'anita.hr','vikas.hr'
Add-ADGroupMember -Identity 'DL_FS_HR_Modify' -Members 'GG_HR_Users'
New-ADGroup -Name 'GG_Finance_Users' -GroupScope Global -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
New-ADGroup -Name 'DL_FS_Finance_Modify' -GroupScope DomainLocal -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
Add-ADGroupMember -Identity 'GG_Finance_Users' -Members 'rohan.fin','neha.fin'
Add-ADGroupMember -Identity 'DL_FS_Finance_Modify' -Members 'GG_Finance_Users'
New-ADGroup -Name 'GG_Operations_Users' -GroupScope Global -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
New-ADGroup -Name 'DL_FS_Operations_Modify' -GroupScope DomainLocal -GroupCategory Security -Path
'OU=Groups,OU=Aster,DC=ad,DC=aster,DC=example'
Add-ADGroupMember -Identity 'GG_Operations_Users' -Members 'meera.ops','arjun.ops'
Add-ADGroupMember -Identity 'DL_FS_Operations_Modify' -Members 'GG_Operations_Users'

```

Line	Command or value	Meaning and parameters
1	Import-Module	Runs the task-specific command with the explicit values shown. Use Get-Help Import-Module -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
2	New-ADGroup	Creates a Security group with the specified Global or DomainLocal scope under the Groups OU.
4	Add-ADGroupMember	Adds only the listed accounts or group to the named group; fresh user logon is needed for reliable access tests.

B Verification and interpretation

```

Get-ADGroupMember GG_HR_Users
Get-ADGroupMember DL_FS_HR_Modify -Recursive

```

Healthy example or criterion	Failed example or criterion
HR recursive membership includes anita.hr and vikas.hr only. Verify equivalent Finance and Operations membership.	A user belongs to the wrong department or the global group is not nested.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
AD group membership is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

D Rollback

Remove only the incorrect membership through Group Properties > Members > select account > Remove. Do not delete a group already used on file ACLs.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B13 Publish three controlled departmental shares

Field	Value
Objective and why	Share and NTFS checks together must grant only the intended departmental access.
Perform on	AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B08 and B12
Evidence produced	EVD-B13 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	FS01 > File Explorer > D:. Create a folder named Departments. Open it.	D:\Departments	Empty parent folder exists.	N
2	Create the department folder. Right-click it > Properties > Security > Advanced. Click Disable inheritance > Remove all inherited permissions from this object.	D:\Departments\HR	Existing inherited user grants are removed; the folder owner can finish this ACL change.	N
3	In Advanced Security Settings click Add > Select a principal. Enter SYSTEM > Check Names > OK. Select Full control and Applies to This folder, subfolders and files. Click OK.	SYSTEM: Full control	SYSTEM permission entry appears.	N
4	Click Add > Select a principal. Enter Administrators > Check Names > OK. Select Full control and This folder, subfolders and files > OK.	BUILTIN\Administrators: Full control	Administrators retain management access.	N
5	Click Add > Select a principal. Enter the resource group > Check Names > OK. Select Modify and This folder, subfolders and files. Click OK > Apply > OK.	ASTER\DL_FS_HR_Modify: Modify	Only the intended group plus SYSTEM and Administrators have grants.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	Folder Properties > Sharing > Advanced Sharing. Tick Share this folder. Enter the hidden share name. Click Permissions.	HR\$	Share permissions window appears.	N
7	Select Everyone and click Remove. Click Add; enter Administrators > Check Names > OK; select Allow Full Control. Click Add again for the department resource group; allow Change and Read. Click OK > Apply > OK > Close.	ASTER\DL_FS_HR_Modify: Change and Read	Share exists with restricted share permissions.	N
8	Create the department folder. Right-click it > Properties > Security > Advanced. Click Disable inheritance > Remove all inherited permissions from this object.	D:\Departments\Finance	Existing inherited user grants are removed; the folder owner can finish this ACL change.	N
9	In Advanced Security Settings click Add > Select a principal. Enter SYSTEM > Check Names > OK. Select Full control and Applies to This folder, subfolders and files. Click OK.	SYSTEM: Full control	SYSTEM permission entry appears.	N
10	Click Add > Select a principal. Enter Administrators > Check Names > OK. Select Full control and This folder, subfolders and files > OK.	BUILTIN\Administrators: Full control	Administrators retain management access.	N
11	Click Add > Select a principal. Enter the resource group > Check Names > OK. Select Modify and This folder, subfolders and files. Click OK > Apply > OK.	ASTER\DL_FS_Finance_Modify: Modify	Only the intended group plus SYSTEM and Administrators have grants.	N
12	Folder Properties > Sharing > Advanced Sharing. Tick Share this folder. Enter the hidden share name. Click Permissions.	Finance\$	Share permissions window appears.	N
13	Select Everyone and click Remove. Click Add; enter Administrators > Check Names > OK; select Allow Full Control. Click Add again for the department resource group; allow Change and Read. Click OK > Apply > OK > Close.	ASTER\DL_FS_Finance_Modify: Change and Read	Share exists with restricted share permissions.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
14	Create the department folder. Right-click it > Properties > Security > Advanced. Click Disable inheritance > Remove all inherited permissions from this object.	D:\Departments\Operations	Existing inherited user grants are removed; the folder owner can finish this ACL change.	N
15	In Advanced Security Settings click Add > Select a principal. Enter SYSTEM > Check Names > OK. Select Full control and Applies to This folder, subfolders and files. Click OK.	SYSTEM: Full control	SYSTEM permission entry appears.	N
16	Click Add > Select a principal. Enter Administrators > Check Names > OK. Select Full control and This folder, subfolders and files > OK.	BUILTIN\Administrators: Full control	Administrators retain management access.	N
17	Click Add > Select a principal. Enter the resource group > Check Names > OK. Select Modify and This folder, subfolders and files. Click OK > Apply > OK.	ASTER\DL_FS_Operations_Modify: Modify	Only the intended group plus SYSTEM and Administrators have grants.	N
18	Folder Properties > Sharing > Advanced Sharing. Tick Share this folder. Enter the hidden share name. Click Permissions.	Operations\$	Share permissions window appears.	N
19	Select Everyone and click Remove. Click Add; enter Administrators > Check Names > OK; select Allow Full Control. Click Add again for the department resource group; allow Change and Read. Click OK > Apply > OK > Close.	ASTER\DL_FS_Operations_Modify: Change and Read	Share exists with restricted share permissions.	Y: EVD-B13

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
New-Item -ItemType Directory -Path 'D:\Departments\HR' -Force
icacls 'D:\Departments\HR' /inheritance:r
```

```

icacls 'D:\Departments\HR' /grant:r '*S-1-5-18:(OI)(CI)F' '*S-1-5-32-544:(OI)(CI)F' 'ASTER\DL_FS_HR_Modify:(OI)(CI)M'
New-SmbShare -Name 'HR$' -Path 'D:\Departments\HR' -FullAccess 'BUILTIN\Administrators' -ChangeAccess
'ASTER\DL_FS_HR_Modify'
New-Item -ItemType Directory -Path 'D:\Departments\Finance' -Force
icacls 'D:\Departments\Finance' /inheritance:r
icacls 'D:\Departments\Finance' /grant:r '*S-1-5-18:(OI)(CI)F' '*S-1-5-32-544:(OI)(CI)F'
'ASTER\DL_FS_Finance_Modify:(OI)(CI)M'
New-SmbShare -Name 'Finance$' -Path 'D:\Departments\Finance' -FullAccess 'BUILTIN\Administrators' -ChangeAccess
'ASTER\DL_FS_Finance_Modify'
New-Item -ItemType Directory -Path 'D:\Departments\Operations' -Force
icacls 'D:\Departments\Operations' /inheritance:r
icacls 'D:\Departments\Operations' /grant:r '*S-1-5-18:(OI)(CI)F' '*S-1-5-32-544:(OI)(CI)F'
'ASTER\DL_FS_Operations_Modify:(OI)(CI)M'
New-SmbShare -Name 'Operations$' -Path 'D:\Departments\Operations' -FullAccess 'BUILTIN\Administrators' -ChangeAccess
'ASTER\DL_FS_Operations_Modify'

```

Line	Command or value	Meaning and parameters
1	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.
2	icacls	Edits or exports NTFS ACLs for the exact path. OI/CI inherit to files/folders; F is Full, M Modify, RX Read/Execute. /inheritance:r removes inherited grants.
4	New-SmbShare	Publishes the exact folder using separate FullAccess, ChangeAccess or ReadAccess share permissions; NTFS must also permit the user.

B Verification and interpretation

```

Get-SmbShare | Where-Object Name -Like "*$"
Get-SmbShareAccess -Name HR$
icacls D:\Departments\HR

```

Healthy example or criterion	Failed example or criterion
HR\$ points to D:\Departments\HR; intended group has Change at share and Modify at NTFS.	Everyone access, wrong path, or missing department group.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied for correct user	Share or NTFS grant missing	Inspect Get-SmbShareAccess and icacls; grant the documented resource group at both layers.
Removed user still has access	Existing logon token or SMB connection	Sign out of the client and sign back in; close prior SMB connections before retesting.
Permission change locks out administrator	Administrators entry omitted	Use the local console as owner/admin; Advanced Security > Change owner to Administrators, then add Full control.

D Rollback

Remove-SmbShare -Name HR\$ removes only the share, not its data. Before changing ACLs, save them using icacls D:\Departments /save C:\MCSA-Evidence\acl-before.txt /t. Restore from the matching parent path; do not grant Everyone as a repair.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

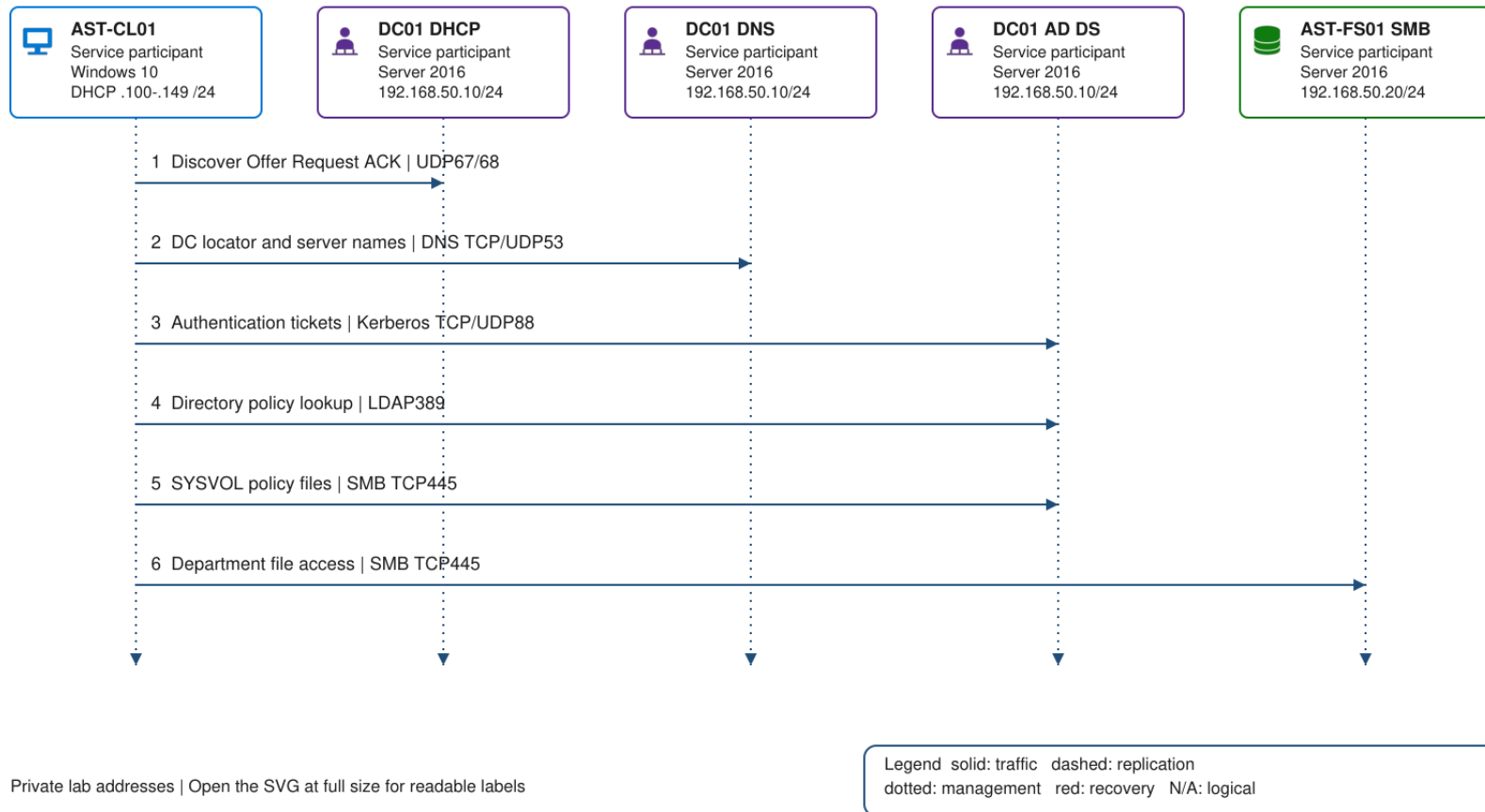
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

PHASE EXIT GATE: run the verification for every task in this phase; record actual output and evidence IDs in the tracker. Continue only when all are correct. Intended groups and both permission layers must be present before user testing.

F6 | Foundation phase E: addressing, policy and acceptance

MCSA-P01 | F6 | Foundation: Client service flow from boot to file access

Version 1.0 | 16 September 2026 | Prepared by _____



Foundation: Client service flow from boot to file access

Current phase: Foundation phase E: addressing, policy and acceptance. Read top to bottom. DHCP supplies addresses and DNS; DNS locates AD; authentication and policy precede the intended share access. These service identities share DC01 in the foundation. The drawing describes conceptual order, not every packet or background renewal.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

B14 Install and configure DHCP

Field	Value
Objective and why	Clients should receive a known address pool and the AD DNS address automatically.
Perform on	AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	45 minutes; add practice and evidence time
Depends on	B07
Evidence produced	EVD-B14 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Open the VMware console for AST-DC01. Sign in with the task administrator account. Open PowerShell and run hostname. Close the console after checking the name.	AST-DC01	The computer name matches the target; no other VM is being changed.	N
2	Start > Server Manager > Manage > Add Roles and Features. On [Before You Begin] click Next.	Run on AST-DC01	[Select installation type] appears.	N
3	[Select installation type]: select Role-based or feature-based installation. Click Next.	Role-based or feature-based installation	[Select destination server] appears.	N
4	[Select destination server]: select Select a server from the server pool. Highlight the named local host. Click Next.	AST-DC01	[Select server roles] appears.	N
5	[Select server roles]: tick DHCP Server. If the required-features dialog opens, leave Include management tools selected and click Add Features.	DHCP Server	The role checkbox is selected.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	On [Select server roles] click Next. On [Select features], keep the automatically selected supporting features and click Next.	Do not select unrelated optional features	The role information page appears, or file services proceed directly to confirmation.	N
7	On [DHCP Server], read the role description and click Next.	No setting is changed on this information page	[Confirm installation selections] appears.	N
8	[Confirm installation selections]: confirm the target hostname and selected roles. Leave Restart the destination server automatically if required cleared. Click Install. On [Installation progress], wait for Installation succeeded on AST-DC01. Click Close.	Only AST-DC01; selected services: DHCP Server	The role and management console are installed. Configuration follows in the next numbered step or task.	N
9	Server Manager notification flag > Complete DHCP configuration. Description > Next. Authorisation: use current ASTER\Administrator credentials > Commit > Close.	Authorise AST-DC01.ad.aster.example	DHCP server is authorised.	N
10	Run dhcpgmt.msc. Expand AST-DC01 > IPv4. Right-click IPv4 > New Scope. Welcome > Next.	Scope name Aster-LAN; description Training client addresses; Next	IP Address Range appears.	N
11	Enter the scope boundaries and subnet mask > Next.	Start 192.168.50.100; End 192.168.50.149; Length 24; mask 255.255.255.0	Add Exclusions and Delay appears.	N
12	Leave exclusions empty because all static servers are outside the scope. Click Next. Set lease duration and click Next.	0 days 8 hours 0 minutes	Configure DHCP Options appears.	N
13	Choose Yes, I want to configure these options now > Next. On Router (Default Gateway), leave the list empty > Next.	No gateway in this isolated subnet	Domain Name and DNS Servers appears.	N
14	Set parent domain and the DNS server. Remove any unintended DNS addresses. Click Next.	ad.aster.example; 192.168.50.10; Add	WINS Servers appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
15	Leave WINS empty > Next. Choose Yes, I want to activate this scope now > Next > Finish.	Active scope	The scope appears under IPv4.	N
16	CL01 > ncpa.cpl > Ethernet > Properties > IPv4 > Properties. Select Obtain an IP address automatically and Obtain DNS server address automatically. Click OK > Close. Open elevated Command Prompt.	ipconfig /release followed by ipconfig /renew	Client gets a .100-.149 address and DNS .10.	Y: EVD-B14

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC -DnsName 'AST-DC01.ad.aster.example' -IPAddress 192.168.50.10
Add-DhcpServerv4Scope -Name 'Aster-LAN' -StartRange 192.168.50.100 -EndRange 192.168.50.149 -SubnetMask 255.255.255.0
    -LeaseDuration 0.08:00:00 -State Active
Set-DhcpServerv4OptionValue -ScopeId 192.168.50.0 -DnsServer 192.168.50.10 -DnsDomain 'ad.aster.example'
# On CL01 only, to replace its temporary static IP:
# Set-NetIPInterface -InterfaceAlias Ethernet -Dhcp Enabled
# Set-DnsClientServerAddress -InterfaceAlias Ethernet -ResetServerAddresses
# ipconfig /renew
```

Line	Command or value	Meaning and parameters
1	Install-WindowsFeature	Installs only the named Windows roles/features; IncludeManagementTools adds the corresponding management consoles.
2	Add-DhcpServerInDC	Runs the task-specific command with the explicit values shown. Use Get-Help Add-DhcpServerInDC -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
3	Add-DhcpServerv4Scope	Runs the task-specific command with the explicit values shown. Use Get-Help Add-DhcpServerv4Scope -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.

Line	Command or value	Meaning and parameters
4	Set-DhcpServerv4OptionValue	Runs the task-specific command with the explicit values shown. Use Get-Help Set-DhcpServerv4OptionValue -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.

B Verification and interpretation

```
Get-DhcpServerv4Scope
Get-DhcpServerv4Lease -ScopeId 192.168.50.0
# On CL01: ipconfig /all
```

Healthy example or criterion	Failed example or criterion
Scope Active; client .100-.149, DHCP server .10, DNS .10.	169.254 address, inactive scope or a lease from VMware.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
APIPA address	No DHCP reply	Confirm VMnet2, authorised service and active scope; renew after fixing.
Wrong DNS option	Option 006 has wrong address	Scope Options > Configure Options > 006 DNS Servers > retain .10 only.
Unauthorised server	Post-install authorisation skipped	DHCP console right-click server > Authorize; refresh after AD is reachable.

D Rollback

For a clean rebuild, deactivate only Aster-LAN; set CL01 temporarily back to 192.168.50.90/24 with DNS .10. Do not remove an active scope needed by later projects.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B15 Set domain account policy and HR user policy

Field	Value
Objective and why	Correct scope is necessary: domain account policy and HR user restrictions solve different problems.
Perform on	AST-DC01 and AST-CL01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	45 minutes; add practice and evidence time
Depends on	B11 and B14
Evidence produced	EVD-B15 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	DC01 > Run gpmmc.msc. Expand Forest > Domains > ad.aster.example. Right-click Default Domain Policy > Edit.	Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy	Domain password policy settings appear.	N
2	Open each named setting. Select Define this policy setting if shown, enter the exact value and click OK.	Enforce password history 5; Maximum password age 90 days; Minimum password age 0 days; Minimum password length 12; Password must meet complexity requirements Enabled; Store passwords using reversible encryption Disabled	All six settings display the lab values.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
3	Open Account Lockout Policy. Set Account lockout threshold first; accept the suggested dependent values, then explicitly set the durations.	Threshold 5; duration 15 minutes; Reset account lockout counter after 15 minutes	Three lockout settings show the required values.	N
4	Close the editor. In GPMC right-click Aster > Users > HR > Create a GPO in this domain, and Link it here.	GPO_HR_ControlPanel_Restriction; OK	New GPO link appears at the HR OU.	N
5	Right-click that GPO > Edit. Open User Configuration > Policies > Administrative Templates > Control Panel. Double-click Prohibit access to Control Panel and PC settings.	Enabled > Apply > OK	User restriction is enabled.	N
6	On CL01 sign in as ASTER\anita.hr. Complete the first-logon password change. Open Command Prompt and run gpupdate /force. Sign out and sign back in.	Run control.exe	Control Panel access is restricted for the HR account.	N
7	Sign out. Sign in as ASTER\rohan.fin and complete the first-logon password change. Run gpupdate /force and control.exe.	Finance comparison	Control Panel opens for Finance.	Y: EVD-B15

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Domain account policy equivalent, run on DC01:
Set-ADDefaultDomainPasswordPolicy -Identity 'ad.aster.example' -ComplexityEnabled $true -MinPasswordLength 12
    -PasswordHistoryCount 5 -MinPasswordAge 0.00:00:00 -MaxPasswordAge 90.00:00:00 -LockoutThreshold 5 -LockoutDuration
    0.00:15:00 -LockoutObservationWindow 0.00:15:00 -ReversibleEncryptionEnabled $false
# This changes AD policy directly. If managing through GPO, keep Default Domain Policy consistent.
New-GPO -Name 'GPO_HR_ControlPanel_Restriction'
New-GPLink -Name 'GPO_HR_ControlPanel_Restriction' -Target 'OU=HR,OU=Users,OU=Aster,DC=ad,DC=aster,DC=example'
Set-GPRegistryValue -Name 'GPO_HR_ControlPanel_Restriction' -Key
    'HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer' -ValueName NoControlPanel -Type DWord -Value 1
```

Line	Command or value	Meaning and parameters
2	Set-ADDefaultDomainPasswordPolicy	Sets the actual default domain account policy. Keep its owning domain GPO consistent so later refreshes do not undo it.
4	New-GPO	Creates a GPO; it will not apply until linked to the intended scope.
5	New-GPLink	Links the named GPO to the exact OU distinguished name; verify the user/computer object belongs there.
6	Set-GPRegistryValue	Writes a policy registry setting into the GPO; HKCU denotes user scope and NoControlPanel value 1 enables the restriction.

B Verification and interpretation

```
Get-ADDefaultDomainPasswordPolicy
# On CL01 under the tested user:
gpresult /r /scope user
gpresult /h "$env:USERPROFILE\Desktop\gpresult.html"
```

Healthy example or criterion	Failed example or criterion
Minimum length 12; threshold 5; HR report includes GPO_HR_ControlPanel_Restriction; Finance report does not.	HR policy denied/not listed; domain account policy still has old values.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
GPO absent	Wrong OU or wrong user context	ADUC: confirm Anita is in HR; log in as Anita before running gpresult.
Finance also restricted	GPO linked at parent/domain	GPMC: remove unintended parent link; keep the HR OU link only; refresh Finance policy.
Password policy not changing	Only an OU policy was edited	Edit the domain-linked account policy and verify Get-ADDefaultDomainPasswordPolicy.

D Rollback

Back up GPOs before later changes. To undo HR restriction, set its user setting to Not Configured and refresh affected users; remove the link only after checking policy result. Record original password settings before changing them.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

B16 Accept the foundation build

Field	Value
Objective and why	A service is accepted only after both allowed and denied paths have been tested.
Perform on	AST-CL01 and AST-DC01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	60 minutes; add practice and evidence time
Depends on	B13 B14 B15
Evidence produced	EVD-B16 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	On CL01 sign in as Anita. File Explorer address bar > enter HR share. Create a text file named hr-check.txt, type HR access verified, save and close.	\\AST-FS01\HR\$	Create/read/delete access works.	N
2	In Explorer enter the Finance and Operations paths separately.	\\AST-FS01\Finance\$ and \\AST-FS01\Operations\$	Access is denied to both. Capture the message, username and path.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
3	Sign out, sign in as Rohan and test all three paths. Create finance-check.txt inside Finance.	Finance allowed; HR and Operations denied	The Finance access row is proven.	N
4	Sign out, sign in as Meera and test all three paths. Create operations-check.txt inside Operations.	Operations allowed; HR and Finance denied	The Operations access row is proven.	N
5	Complete the same three-path test after signing in separately as Vikas, Neha and Arjun. Record each result separately in the supplied Tests sheet.	18 access outcomes across six users	No test relies on another user's cached SMB session.	N
6	On DC01 create C:\MCSA-Evidence in File Explorer. Run the export commands below from elevated PowerShell.	Attach exports to the final report	Inventory and policy evidence is saved without passwords.	Y: EVD-B16

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Verification task: execute these read-only checks; no new configuration is required.
New-Item C:\MCSA-Evidence -ItemType Directory -Force
Get-ADUser -Filter * -SearchBase 'OU=Users,OU=Aster,DC=ad,DC=aster,DC=example' | Select-Object SamAccountName,Enabled |
    Export-Csv C:\MCSA-Evidence\users.csv -NoTypeInfoation
Get-DhcpServerv4Scope | Export-Csv C:\MCSA-Evidence\scopes.csv -NoTypeInfoation
Get-ADDefaultDomainPasswordPolicy | Format-List | Out-File C:\MCSA-Evidence\password-policy.txt
```

Line	Command or value	Meaning and parameters
2	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.
3	Get-ADUser	Reads the current identity/configuration or tests the named endpoint. Interpret the output using this task's healthy and failed examples; do not treat output absence as automatic success.

B Verification and interpretation

```
# CL01 user context:
whoami
ipconfig /all
nslookup AST-FS01.ad.aster.example
gpresult /r
```

Healthy example or criterion	Failed example or criterion
Correct identity, DHCP lease, DNS answer and policy; all 18 access outcomes match the matrix.	Any unauthorised successful access or a legitimate department blocked.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
All shares open as admin	Testing privileged account	Sign out completely and retest as each standard user.
Multiple credentials conflict	Prior SMB connection remains	Use a fresh logon session; do not mix different users in one Explorer session.
Export path missing	Evidence directory not created	Create C:\MCSA-Evidence before rerunning the export command.

D Rollback

Delete only the three disposable check files after evidence collection. Preserve account, share and policy configuration as the accepted baseline.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

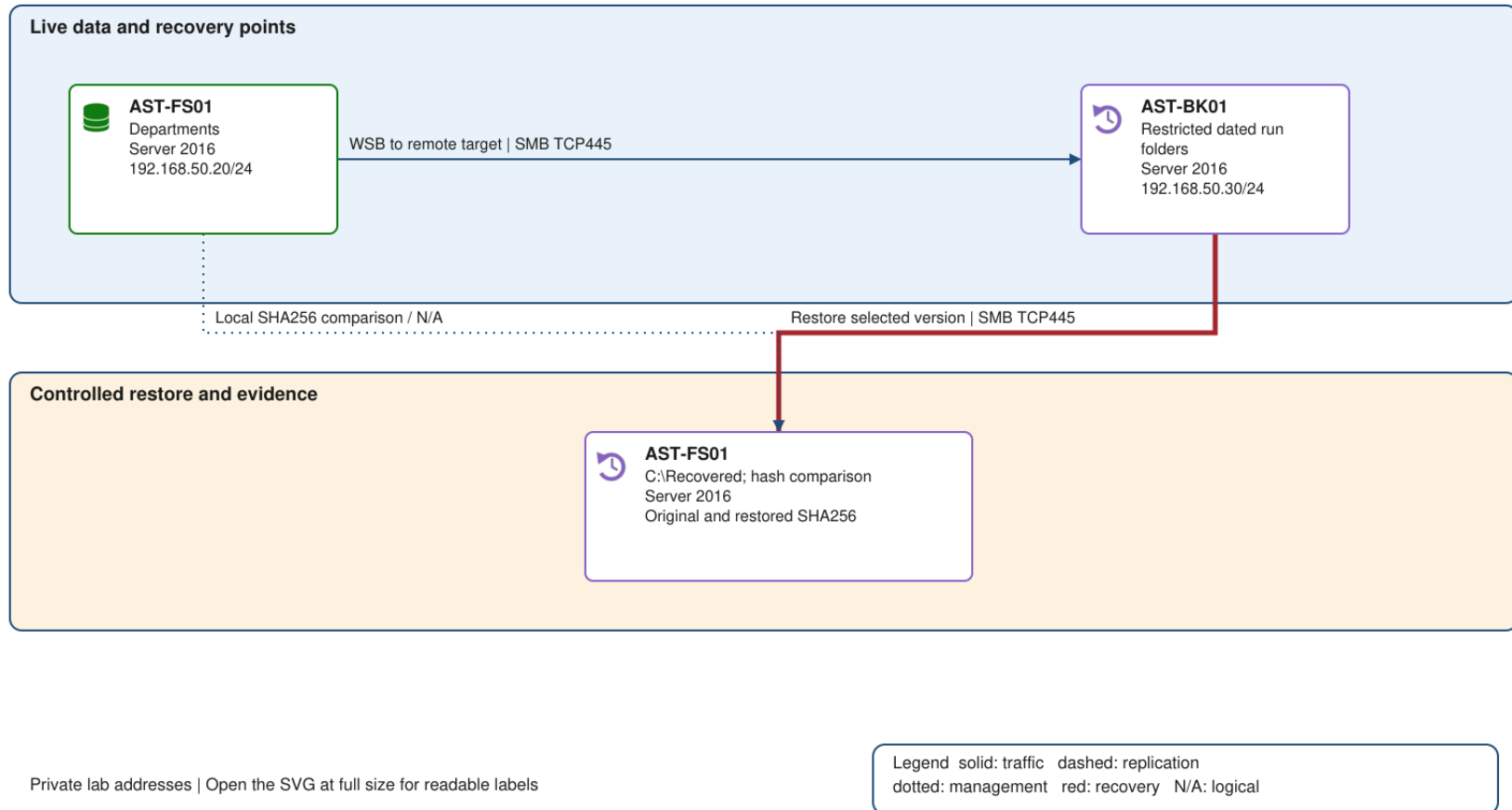
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

PHASE EXIT GATE: run the verification for every task in this phase; record actual output and evidence IDs in the tracker. Continue only when all are correct. B16 requires 18 departmental access outcomes and the correct HR-only policy scope.

D8 | Specialist phase: Backup destination and first recovery point

MCSA-P01 | D8 | Backup and verified file restoration

Version 1.0 | 16 September 2026 | Prepared by _____



Backup and verified file restoration

Current phase: Specialist phase: Backup destination and first recovery point. The backup travels to a restricted share; restoration selects a particular completed version and writes to another location. Compare its hash before returning the file to service. Separate run directories preserve distinct test recovery points. This same-host lab does not provide off-site protection.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

P1-01 Prepare a restricted backup destination

Field	Value
Objective and why	The backup target needs a dedicated identity and restricted storage before a job can run.
Perform on	AST-BK01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	35 minutes; add practice and evidence time
Depends on	B16
Evidence produced	EVD-P1-01 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	BK01 > Run compmgmt.msc > Local Users and Groups > Users. Right-click Users > New User.	User name backupwriter; unique lab-only password; clear User must change password at next logon; Create > Close	Local non-administrator backupwriter exists.	N
2	File Explorer > D:.. Create Backup and then create Run01 inside Backup. Right-click Backup > Properties > Security > Advanced > Disable inheritance > Remove all inherited permissions.	D:\Backup\Run01	Backup folder is ready for restricted ACLs.	N
3	Add SYSTEM and Administrators with Full control. Add AST-BK01\backupwriter with Modify. For each: Add > Select a principal > type name > Check Names > OK > permission > This folder, subfolders and files > OK.	Three explicit entries; Apply > OK	Backup writer can write without local administrator membership.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
4	Backup Properties > Sharing > Advanced Sharing > Share this folder. Enter Backup\$ > Permissions. Remove Everyone; add Administrators Full Control and backupwriter Change/Read.	Share \\AST-BK01\Backup\$	Restricted backup share exists.	N
5	FS01 > elevated Command Prompt. Test the share with explicit backup credentials; enter the password only at the prompt.	net use \\AST-BK01\Backup\$ /user:AST-BK01\backupwriter *	The command completed successfully. Do not screenshot the password.	Y: EVD-P1-01

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

On BK01 only:

```
New-LocalUser -Name backupwriter -Password (Read-Host 'Backup writer password' -AsSecureString)
New-Item D:\Backup\Run01 -ItemType Directory -Force
icacls D:\Backup /inheritance:r
icacls D:\Backup /grant:r '*S-1-5-18:(OI)(CI)F' '*S-1-5-32-544:(OI)(CI)F' 'AST-BK01\backupwriter:(OI)(CI)M'
New-SmbShare -Name 'Backup$' -Path D:\Backup -FullAccess 'BUILTIN\Administrators' -ChangeAccess 'AST-BK01\backupwriter'
```

Line	Command or value	Meaning and parameters
2	New-LocalUser	Runs the task-specific command with the explicit values shown. Use Get-Help New-LocalUser -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.
3	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.
4	icacls	Edits or exports NTFS ACLs for the exact path. OI/CI inherit to files/folders; F is Full, M Modify, RX Read/Execute. /inheritance:r removes inherited grants.
6	New-SmbShare	Publishes the exact folder using separate FullAccess, ChangeAccess or ReadAccess share permissions; NTFS must also permit the user.

B Verification and interpretation

```
Get-SmbShareAccess -Name 'Backup$'  
Get-LocalGroupMember Administrators  
# FS01: Test-Path '\\AST-BK01\Backup$\Run01'
```

Healthy example or criterion	Failed example or criterion
Backup share grants Administrators and backupwriter only; backupwriter is not an administrator; FS01 can reach Run01.	Access denied, wrong target name, or backupwriter in Administrators.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
Backup share is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

D Rollback

Disconnect only the test SMB session using net use \\AST-BK01\Backup\$ /delete. If rebuilding, remove the share after preserving any backups; never delete D:\Backup before inspecting its contents.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

P1-02 Create test data and the first backup

Field	Value
Objective and why	Record the original content before backing it up so restoration can be checked independently.
Perform on	AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	40 minutes; add practice and evidence time
Depends on	P1-01
Evidence produced	EVD-P1-02 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	FS01 > elevated PowerShell. Create evidence and seed a small document. Run the commands in the equivalent block through Get-FileHash.	D:\Departments\HR\restore-demo.txt	The file and its SHA256 record exist.	N
2	Server Manager > Manage > Add Roles and Features > Before You Begin Next > Role-based installation Next > local FS01 Next > Select server roles Next. On Select features tick Windows Server Backup > Next > Install > Close.	Windows Server Backup feature	Windows Server Backup console is available.	N
3	Run wbadmin.msc. Select Local Backup. Actions > Backup Once. On Backup Options select Different options > Next.	Custom backup	Select Backup Configuration appears.	N
4	Select Custom > Next. Select Items for Backup > Add Items. Expand D: and select Departments only > OK > Next.	D:\Departments	Specify Destination Type appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Choose Remote shared folder > Next. Type the exact destination. Use the default access-control choice for this isolated exercise; the destination share is already restricted.	\\AST-BK01\Backup\$\Run01	Credentials prompt appears when needed.	N
6	Enter AST-BK01\backupwriter and its password. Review the source and destination on Confirmation; click Backup. Wait for Completed > Close.	Back up Departments only	Backup result is successful; record start and finish time.	Y: EVD-P1-02

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
New-Item C:\MCSA-Evidence -ItemType Directory -Force
Set-Content D:\Departments\HR\restore-demo.txt 'Aster approved procedure version 1'
Get-FileHash D:\Departments\HR\restore-demo.txt -Algorithm SHA256 | Export-Csv C:\MCSA-Evidence\original-hash.csv
-NoTypeInfo
Install-WindowsFeature Windows-Server-Backup
# Establish the restricted SMB session from P1-01, then run:
wbadmin start backup -backupTarget:\\AST-BK01\Backup$\Run01 -include:D:\Departments -vssCopy
# Review the confirmation. Do not embed passwords in a script.
```

Line	Command or value	Meaning and parameters
1	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.
2	Set-Content	Writes the exact harmless test text to the selected file; an existing file at that path is replaced.
3	Get-FileHash	Computes an integrity digest; compare with the value recorded before the file was deleted.
4	Install-WindowsFeature	Installs only the named Windows roles/features; IncludeManagementTools adds the corresponding management consoles.

Line	Command or value	Meaning and parameters
6	wbadmin	Runs native Windows backup/recovery. backupTarget identifies a specific run folder; include/items identify data; get versions is read-only.

B Verification and interpretation

wbadmin get versions -backupTarget:\\AST-BK01\Backup\$\Run01
Get-WBSummary

Healthy example or criterion	Failed example or criterion
A successful backup version with a date/time is listed. Record the actual identifier; it is generated by this build.	No backup found or LastBackupResultHR indicates a failure; inspect WSB operation result.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Backup target denied	Writer identity or ACL wrong	Retest the P1-01 SMB session; verify both share and NTFS permissions.
No backup version	Job incomplete or wrong subfolder queried	Use exactly Run01 and wait for Completed before enumerating versions.
Insufficient disk space	Backup destination is full	Check BK01 D: free space; preserve the previous verified backup before clearing any old run.

D Rollback

Cancel an incomplete job in the console or use wbadmin stop job after reviewing its prompt. Remove an incomplete Run01 only after confirming no valid restore point exists there.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

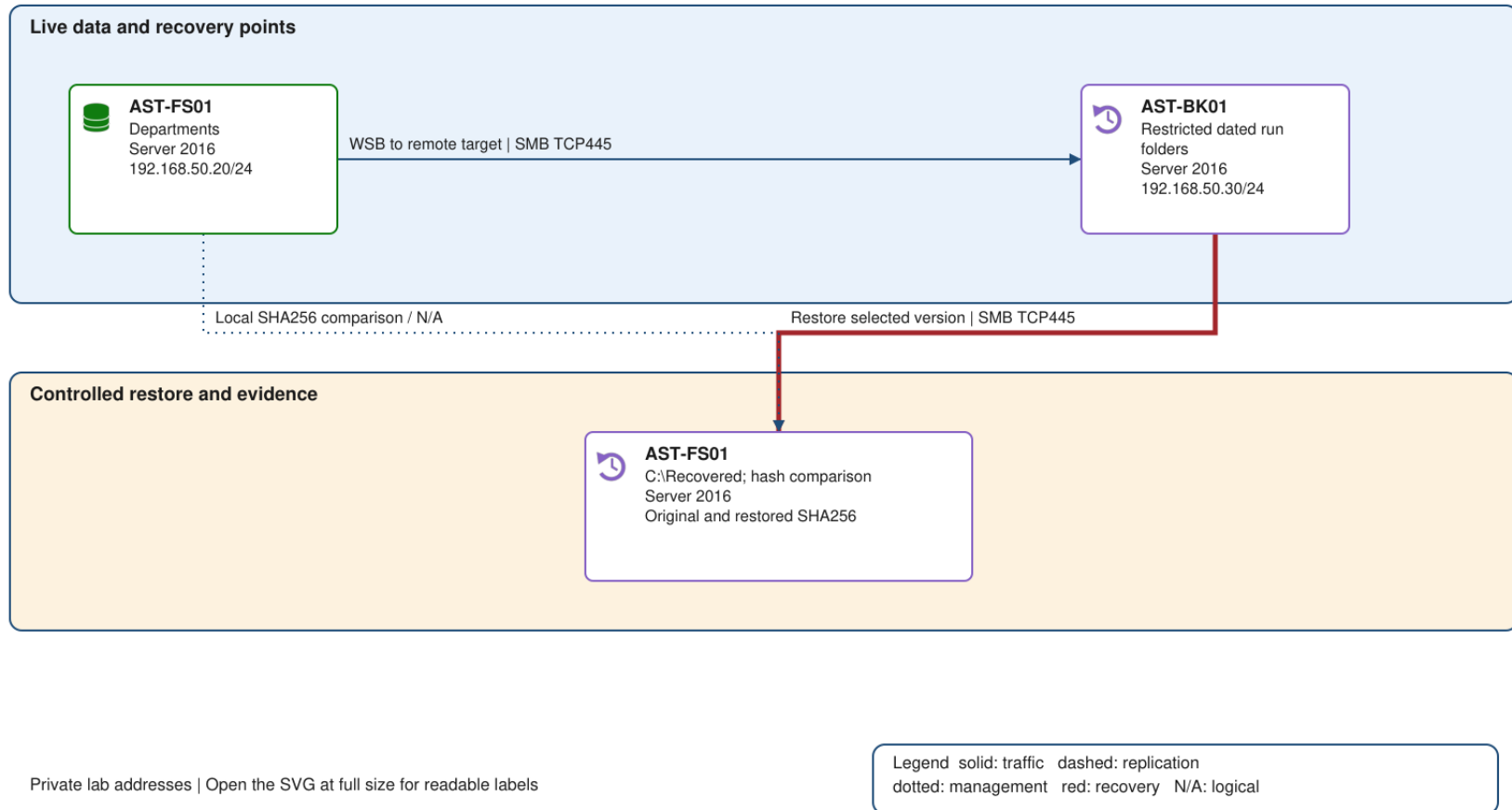
Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames

SPECIALIST PHASE EXIT GATE: complete the task verification above, preserve the evidence and confirm rollback/recovery instructions are understood. Do not accept an installed role as proof of a working service. Return every intentionally changed service or account to the documented state before the next phase.

D8 | Specialist phase: Restore, integrity and retention

MCSA-P01 | D8 | Backup and verified file restoration

Version 1.0 | 16 September 2026 | Prepared by _____



Backup and verified file restoration

Current phase: Specialist phase: Restore, integrity and retention. The backup travels to a restricted share; restoration selects a particular completed version and writes to another location. Compare its hash before returning the file to service. Separate run directories preserve distinct test recovery points. This same-host lab does not provide off-site protection.

Read the named service path before starting the tasks below. Compare the VM name, network and IP with the build sheet. The gate at the end of this phase must pass before moving on.

P1-03 Delete and restore a file with integrity proof

Field	Value
Objective and why	Recovery, not job completion, establishes that the backup is useful.
Perform on	AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	40 minutes; add practice and evidence time
Depends on	P1-02
Evidence produced	EVD-P1-03 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	Confirm original-hash.csv exists and wbadmin lists the completed Run01 backup. Note the current time in the logbook.	Recovery timer starts now	A recovery point and independent comparison value exist.	N
2	File Explorer > D:\Departments\HR. Delete restore-demo.txt only. Confirm its path is absent.	Never delete the department folder	The working copy is gone.	N
3	Open Windows Server Backup > Local Backup > Recover. On Getting Started choose A backup stored on another location > Next.	Remote shared folder > Next	Specify Remote Folder appears.	N
4	Enter the completed run path and authenticate when asked. Click Next. Select the actual backup date and time > Next.	\\AST-BK01\Backup\$\Run01	Select Recovery Type appears.	N
5	Select Files and folders > Next. Expand D:\Departments\HR and select restore-demo.txt > Next.	One test file only	Specify Recovery Options appears.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
6	Select Another location. Create C:\Recovered in Explorer and use it as the destination. Retain security permissions when offered. Click Next > Recover.	C:\Recovered	Recovery completes without overwriting live departmental files.	N
7	Find the recovered file below C:\Recovered; WSB may recreate path components. Run the comparison block. Record finish time and elapsed minutes.	Exactly one restored restore-demo.txt	Original and restored SHA256 values match. Copy the verified file to its original folder only after this check.	Y: EVD-P1-03

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# Read the recovery points and their actual version identifiers:
wbadmin get versions -backupTarget:\\AST-BK01\Backup$\Run01
$version=Read-Host 'Paste the Version identifier shown above'
New-Item C:\Recovered -ItemType Directory -Force
wbadmin start recovery -version:$version -itemType:File -items:D:\Departments\HR\restore-demo.txt
    -backupTarget:\\AST-BK01\Backup$\Run01 -recoveryTarget:C:\Recovered
$restored=Get-ChildItem C:\Recovered -Recurse -Filter restore-demo.txt -File
if (@($restored).Count -ne 1) { throw 'Expected exactly one recovered file' }
$expected=(Import-Csv C:\MCSA-Evidence\original-hash.csv).Hash
$actual=(Get-FileHash $restored.FullName -Algorithm SHA256).Hash
$expected -eq $actual
```

Line	Command or value	Meaning and parameters
2	wbadmin	Runs native Windows backup/recovery. backupTarget identifies a specific run folder; include/items identify data; get versions is read-only.
3	\$version=Read-Host	Stores a named value or result used by a later line. Read the assigned path/value before execution; a prompt must use your actual observed result.
4	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.

Line	Command or value	Meaning and parameters
7	if	Runs the task-specific command with the explicit values shown. Use Get-Help if -Full for local syntax if your installed build rejects a parameter; do not substitute unrelated settings.

B Verification and interpretation

Run the final hash comparison above and read the recovered file in Notepad.

Healthy example or criterion	Failed example or criterion
True; original text preserved; elapsed recovery time recorded, target <=30 minutes.	False, no restored file, or multiple ambiguous restored copies.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Wrong backup date	Restore point selected before the test file existed	Choose the Run01 version created after seeding the file.
File seems missing	Restore preserves subfolder structure	Search C:\Recovered recursively for restore-demo.txt.
Hash mismatch	Wrong version or modified file	Do not declare pass; compare backup date and expected hash, then restore the correct recovery point.

D Rollback

The restore uses another location to preserve current data. If it is the wrong version, remove only that recovered test file and repeat. Restore the accepted file to HR after verification.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

P1-04 Prove retention and record the operational schedule

Field	Value
Objective and why	Two independent run folders make recovery-point choice observable and avoid silently overwriting the first backup.
Perform on	AST-BK01 and AST-FS01
Logged in as	ASTER\Administrator unless a step names a standard user; before domain creation/join use the guest local Administrator or labadmin. Host actions use the host administrator.
Estimated guided time	30 minutes; add practice and evidence time
Depends on	P1-03
Evidence produced	EVD-P1-04 screenshot(s), actual verification output and logbook entry

Step #	Exact action and navigation	Exact value	Expected result	Capture
1	On BK01 create D:\Backup\Run02 in File Explorer. On FS01 change restore-demo.txt to version 2 using Notepad and save.	Aster approved procedure version 2	The working file differs from the Run01 state.	N
2	FS01 > wbadmin.msc > Backup Once > Different options > Next > Custom > Next > Add Items > D:\Departments > OK > Next.	Same source as Run01	Remote shared folder destination page can be selected.	N
3	Choose Remote shared folder > Next, enter Run02, supply backupwriter credentials when requested, review and click Backup.	\\AST-BK01\Backup\$\Run02	Second backup completes without writing into Run01.	N
4	Run wbadmin get versions against Run01 and Run02 separately. Record the version identifier, target and file version in the logbook.	Two targets, two independently listed recovery points	Both runs are retained.	N

Step #	Exact action and navigation	Exact value	Expected result	Capture
5	Record the lab operator calendar in the change template: daily 17:00 manual backup while VMs are running; use a new dated destination; daily result check; weekly restore test; retain seven successful daily run folders only when storage permits.	This exercise implements manual runs, not unattended scheduling	The handover states who runs the backup and how missed runs are escalated.	Y: EVD-P1-04

A PowerShell or native command equivalent

The numbered steps define the complete task. Use the GUI or the command block for each matching configuration operation; do not create the same object twice. Complete the remaining manual actions, sign-ins, fault injections and restoration steps even when you use PowerShell. Host comments identify where each command runs. Copy long commands from HTML or the reference file; printed lines may wrap.

```
# On BK01:
New-Item D:\Backup\Run02 -ItemType Directory -Force
# On FS01:
Set-Content D:\Departments\HR\restore-demo.txt 'Aster approved procedure version 2'
wbadmin start backup -backupTarget:\\AST-BK01\Backup$\Run02 -include:D:\Departments -vssCopy
wbadmin get versions -backupTarget:\\AST-BK01\Backup$\Run01
wbadmin get versions -backupTarget:\\AST-BK01\Backup$\Run02
```

Line	Command or value	Meaning and parameters
2	New-Item	Creates the named folder(s); Force permits reuse of an existing folder but does not validate its ACL.
4	Set-Content	Writes the exact harmless test text to the selected file; an existing file at that path is replaced.
5	wbadmin	Runs native Windows backup/recovery. backupTarget identifies a specific run folder; include/items identify data; get versions is read-only.

B Verification and interpretation

Enumerate both destinations; inspect free disk space on BK01 with Get-Volume -DriveLetter D.

Healthy example or criterion	Failed example or criterion
Both run folders contain usable versions; backup owner and missed-run action documented.	Only Run02 exists or retention is claimed without preserved backups.

Examples are expected criteria, not captured results from a running Windows lab. Write your actual output and timestamp in the evidence tracker. Stop before the next dependent task if the criterion is not met.

C Common errors

Symptom	Cause	Fix
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
Backup retention is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

D Rollback

Do not delete Run01 until Run02 has been restored and checked. Capacity cleanup requires recording which recovery point is being removed. A schedule document alone is not evidence of a completed backup.

E Server version note

Windows Server 2016 Desktop Experience is the reference. Newer Server Manager screenshots may differ; use the named console and setting. A task marked VERIFY IN YOUR BUILD requires the local help check stated in that task.

Trainee observation _____ Actual minutes _____ Issue and fix _____ Evidence filenames _____

SPECIALIST PHASE EXIT GATE: complete the task verification above, preserve the evidence and confirm rollback/recovery instructions are understood. Do not accept an installed role as proof of a working service. Return every intentionally changed service or account to the documented state before the next phase.

SECTION 6 EVIDENCE AND DOCUMENTATION PACK

Evidence ID	Task	Exactly what to show	Filename	Report location
EVD-P1-01	P1-01	Backup share grants Administrators and backupwriter only; backupwriter is not an administrator; FS01 can reach Run01.	EVD-P1-01_verified.png	Implementation and results
EVD-P1-02	P1-02	A successful backup version with a date/time is listed. Record the actual identifier; it is generated by this build.	EVD-P1-02_verified.png	Implementation and results
EVD-P1-03	P1-03	True; original text preserved; elapsed recovery time recorded, target <=30 minutes.	EVD-P1-03_verified.png	Implementation and results
EVD-P1-04	P1-04	Both run folders contain usable versions; backup owner and missed-run action documented.	EVD-P1-04_verified.png	Implementation and results

Capture hostname, signed-in identity where relevant, the exact setting or command and its result. Use additional suffixes _a, _b when one screenshot cannot show both configuration and verification. Keep passwords and shared secrets outside the crop. Expected-output examples in the guide cannot be submitted as actual evidence.

Logbook field	What to enter
Date/time and task	Local date/time, timezone, task ID and host
Effort	Actual minutes, not the estimated allowance
Issue	Exact symptom and error code
Diagnosis	Commands and observations used to isolate the cause
Resolution	Specific setting changed and successful retest
Learning	Why the setting mattered and what you would check first next time

Change record	Required entries
CR identifier	Project, date, owner and reviewer
Before	Existing setting and exported baseline
Plan	Why, hosts affected, dependencies, steps and expected impact
Rollback	Exact reversal, trigger for rollback and responsible person
After	Actual result, related test IDs, evidence and reviewer acceptance

Risk	Mitigation
Wrong VM or disk changed	Check hostname, disk size and task ID before each mutation
Lost restore point	Preserve last verified backup until a newer one is restored
Unclear fault cause	Inject one fault at a time and restore accepted state
Insufficient host capacity	Use the active-VM schedule and monitor free storage
Credentials in evidence	Use private prompts and crop secrets before submission

Configuration exports: run the read-only verification commands for every task and save outputs. The supplied Scripts/ReadOnly_Checks.ps1 performs a small role-aware local inventory. This workbook includes AD, DHCP and policy exports; Project 6 additionally includes GPO and firewall baselines. Do not fabricate exports for roles absent from this project.

Final report order: title and identity; project charter; business why; design decisions; final architecture; VM/IP inventory; task evidence; actual test results; five incident and recovery records; operations handover; measured results; limitations; sources; sign-off. Use Templates/Project_Report_Template.docx and the editable tracker.

Foundation evidence	Capture and report requirement
EVD-B01	VMnet2 host-only; VMware DHCP disabled; 16 GB host RAM and adequate free SSD space.
EVD-B02	AST-DC01; IPv4 192.168.50.10; DNS 192.168.50.10; NTFS data volume when specified.
EVD-B03	AST-FS01; IPv4 192.168.50.20; DNS 192.168.50.10; NTFS data volume when specified.
EVD-B04	AST-BK01; IPv4 192.168.50.30; DNS 192.168.50.10; NTFS data volume when specified.

Foundation evidence	Capture and report requirement
EVD-B05	AST-CL01; IPv4 192.168.50.90; DNS 192.168.50.10; NTFS data volume when specified.
EVD-B06	Install State: Installed. The server is not yet a domain controller.
EVD-B07	DNSRoot ad.aster.example; SYSVOL and NETLOGON shares; LDAP SRV answer identifies AST-DC01 on port 389.
EVD-B08	Domain ad.aster.example; nltest returns a DC name and successful status.
EVD-B09	Domain ad.aster.example; nltest returns a DC name and successful status.
EVD-B10	Domain ad.aster.example; nltest returns a DC name and successful status.
EVD-B11	Six departmental accounts enabled; three member computer objects in the documented OUs.
EVD-B12	HR recursive membership includes anita.hr and vikas.hr only. Verify equivalent Finance and Operations membership.
EVD-B13	HR\$ points to D:\Departments\HR; intended group has Change at share and Modify at NTFS.
EVD-B14	Scope Active; client .100-.149, DHCP server .10, DNS .10.
EVD-B15	Minimum length 12; threshold 5; HR report includes GPO_HR_ControlPanel_Restriction; Finance report does not.
EVD-B16	Correct identity, DHCP lease, DNS answer and policy; all 18 access outcomes match the matrix.

SECTION 7 VALIDATION AND TESTING

TC ID	Objective	Precondition	Test action	Expected result
TC-1-01	Prepare a restricted backup destination	B16	Run task B verification on AST-BK01	Backup share grants Administrators and backupwriter only; backupwriter is not an administrator; FS01 can reach Run01.
TC-1-02	Create test data and the first backup	P1-01	Run task B verification on AST-FS01	A successful backup version with a date/time is listed. Record the actual identifier; it is generated by this build.
TC-1-03	Delete and restore a file with integrity proof	P1-02	Run task B verification on AST-FS01	True; original text preserved; elapsed recovery time recorded, target <=30 minutes.
TC-1-04	Prove retention and record the operational schedule	P1-03	Run task B verification on AST-BK01 and AST-FS01	Both run folders contain usable versions; backup owner and missed-run action documented.

The editable Evidence_and_Evaluation.xlsx Tests sheet includes Category, Actual result, Pass/Fail and Evidence ID fields. Initial status is Not run. Add the 18 departmental outcomes and five drill records; blank observed fields are intentional trainee input, not completed evidence.

7.2 Command verification pack

P1-01 verification

```
Get-SmbShareAccess -Name 'Backup$'
Get-LocalGroupMember Administrators
# FS01: Test-Path '\\AST-BK01\Backup$\Run01'
```

Healthy	Failed	Interpretation
Backup share grants Administrators and backupwriter only; backupwriter is not an administrator; FS01 can reach Run01.	Access denied, wrong target name, or backupwriter in Administrators.	An actual mismatch blocks acceptance; diagnose the named task before moving forward.

P1-02 verification

wbadmin get versions -backupTarget:\\AST-BK01\Backup\$\Run01
Get-WBSummary

Healthy	Failed	Interpretation
A successful backup version with a date/time is listed. Record the actual identifier; it is generated by this build.	No backup found or LastBackupResultHR indicates a failure; inspect WSB operation result.	An actual mismatch blocks acceptance; diagnose the named task before moving forward.

P1-03 verification

Run the final hash comparison above and read the recovered file in Notepad.

Healthy	Failed	Interpretation
True; original text preserved; elapsed recovery time recorded, target <=30 minutes.	False, no restored file, or multiple ambiguous restored copies.	An actual mismatch blocks acceptance; diagnose the named task before moving forward.

P1-04 verification

Enumerate both destinations; inspect free disk space on BK01 with Get-Volume -DriveLetter D.

Healthy	Failed	Interpretation
Both run folders contain usable versions; backup owner and missed-run action documented.	Only Run02 exists or retention is claimed without preserved backups.	An actual mismatch blocks acceptance; diagnose the named task before moving forward.

7.3 Controlled destructive drills

Only disposable lab objects may be changed. Read the restoration instructions before injecting each fault. Retain a working administrator console and a verified backup where deletion is involved. Return to the accepted state between drills.

DR-1-01 Wrong client DNS

Field	Execution
Perform on	CL01
Precondition	B16 accepted; note DHCP DNS configuration
What and how to break	ncpa.cpl > Ethernet > IPv4 > use DNS 192.168.50.254; ipconfig /flushdns.
What should still work	IP connectivity to FS01 may remain; fresh domain DNS queries should fail.
Proof	nslookup AST-FS01.ad.aster.example must not resolve through .254; compare Test-NetConnection 192.168.50.20 -Port 445.
Restore and retest	Return DNS to Obtain automatically; ipconfig /renew; ipconfig /flushdns; retest resolution.

DR-1-02 DHCP stopped

Field	Execution
Perform on	DC01 and CL01
Precondition	Record client lease and have VM consoles open
What and how to break	DC01 services.msc > DHCP Server > Stop. CL01 ipconfig /release then ipconfig /renew.
What should still work	Static servers remain reachable to each other; client cannot obtain a new valid DHCP lease.
Proof	Capture renewal failure and DC static address; existing lease survival alone is not the test.
Restore and retest	Start DHCP Server on DC01; CL01 ipconfig /renew; verify scope address and DNS.

DR-1-03 HR GPO link disabled

Field	Execution
Perform on	DC01 and CL01
Precondition	Capture baseline Anita policy
What and how to break	GPMC > HR OU > right-click GPO_HR_ControlPanel_Restriction link > clear Link Enabled.
What should still work	Other domain services remain; HR restriction may disappear after refresh and sign-in.
Proof	Anita gpupdate /force; sign out/in; gpresult /r and control.exe.
Restore and retest	Re-enable only the HR link; refresh and sign in; restriction and gpresult must return.

DR-1-04 File service stopped

Field	Execution
Perform on	FS01 and CL01
Precondition	Close active test files and capture share access
What and how to break	FS01 services.msc > Server > Stop; acknowledge dependent-service stop only in this disposable lab.
What should still work	Domain authentication and DNS still work; new SMB sessions fail.
Proof	Fresh CL01 Test-NetConnection AST-FS01 -Port 445 and share-open attempt.
Restore and retest	Start Server service; restart any dependent service stopped by the prompt; verify shares and ACL results.

DR-1-05 Delete and restore the test file

Field	Execution
Perform on	FS01
Precondition	P1-02 completed backup and saved original hash
What and how to break	Delete only restore-demo.txt as specified in P1-03.
What should still work	The backup remains independently available on BK01.
Proof	Execute every P1-03 recovery step; matching hash and recorded elapsed time are required.
Restore and retest	Return the verified restored file to its original folder; preserve the recovery evidence.

7.4 Security validation

Control	Objective proof
Least privilege	Departmental account succeeds only on intended resource; admin accounts are not used as substitutes
Password policy	Domain policy export and disposable-account weak-password rejection; lockout test where included
Audit trail	Relevant service/backup/security event with host and time; file SACL test only where implemented
Firewall	Relevant service test under an enabled firewall; fault removal restores the named path
Secrets	No password, DSRM value, shared key or licence key in evidence

Acceptance thresholds: every mandatory functional test passes; all five faults are restored; no unexplained cross-department access; backup projects show a matching restored hash and a measured single-file restore target of 30 minutes; HA project demonstrates a never-before-used user logon and a fresh lease during outage; remaining defects are resolved or explicitly marked incomplete. No arbitrary performance benchmark is claimed for a single laptop.

SECTION 8 GO LIVE READINESS AND DEFINITION OF DONE

Go-live here means lab acceptance and handover, not production release. The project is complete only when the following records exist.

Check	Evidence or sign-off
☐ Foundation accepted	B16 and all 18 access outcomes
☐ All project task checks passed	Each TC row has actual result, status and matching evidence
☐ Five drills completed and restored	DR records with before/failure/recovery
☐ Architecture matches actual build	Correct guest names, IPs, roles and failure boundaries
☐ Backup/recovery claim limited to what was tested	Actual restore points and measured result where applicable
☐ No pending faults or orphaned broad access	Membership, service and firewall review
☐ Handover demonstrated	A second trainee reproduces three checks
☐ Interview defence and live change completed	Trainer rubric and feedback

Prepared by	Reviewed by	Approved for lab completion	Date
_____	_____	_____	_____

SECTION 9 TROUBLESHOOTING RUNBOOK

Use the first failing dependency: virtual network and address, then DNS/time, then identity, then service, then permissions or application settings. Compare local versus remote behaviour. The foundation cases below are supplemented by at least three errors under every project task.

Symptom	Cause	Diagnostic	Step-by-step repair	Prevention
Domain not found	Member DNS points to router/public resolver	ipconfig /all; nslookup -type=SRV _ldap._tcp.dc._msdc.s.ad.aster.example	ncpa.cpl > IPv4 > preferred DNS .10; clear wrong alternates; ipconfig /flushdns; repeat join precheck.	Keep AD DNS on domain members.
169.254 address	No usable DHCP offer	ipconfig /all; Get-DhcpServerv4Scope	Check VMnet2, DHCP authorisation, active scope and free leases; then ipconfig /renew.	Disable VMware DHCP and use one documented pool.
Wrong lease/network	NIC attached to NAT or another VMnet	ipconfig /all	Power off guest; VM Settings > Network Adapter > Custom VMnet2; reconnect and renew.	Check vNIC placement before service debugging.
Duplicate IP warning	Static address overlaps another endpoint	Get-NetIPAddress; arp -a	Compare inventory and both VM consoles; correct only the duplicate machine; renew dynamic clients.	Keep static .10-.90 outside .100-.149.
Server Core command prompt	Wrong installation image selected	Get-ComputerInfo Select WindowsProductName	Rebuild the empty training guest with Desktop Experience; preserve any evidence first.	Check edition before clicking Install.
D drive missing	DVD owns D or disk is uninitialised	Get-Disk; Get-Volume	diskmgmt.msc: move DVD letter to Z; initialise verified empty secondary disk GPT; create NTFS D.	Check size and disk number before formatting.
Share access denied	Share or NTFS layer rejects request	Get-SmbShareAccess HR\$; icacls D:\Departments\HR	Check resource group at both layers; correct intended group grant; sign out/in before retest.	Test standard accounts and denied paths.
Removed user retains access	Old access token or SMB session	whoami /groups; net use	Close sessions, sign out completely and sign back in; test in a new session.	Record the tested identity and time.

Symptom	Cause	Diagnostic	Step-by-step repair	Prevention
Everyone can read HR	Broad inherited or share grant	icacls D:\Departments\HR	Save ACL; remove only the unintended user grant; retain SYSTEM/admin; run all department tests.	Use the approved access matrix.
GPO missing from report	Wrong OU, absent link or security filtering	gpresult /r /scope user	Check the user object OU, link enabled state and Read/Apply permissions; refresh and sign in again.	Match policy type to object scope.
GPO scope surprising	Block inheritance, enforced link, loopback or WMI filter changes evaluation	gpresult /h C:\MCSA-Evidence\policy.html	GPMC: inspect inheritance and link; inspect GPO WMI Filtering; for this baseline use None. Check computer loopback policy; restore Not Configured unless the project explicitly needs it.	Do not copy unexplained filters from other tutorials.
OU password setting ineffective	Domain accounts use domain policy or a valid fine-grained policy	Get-ADDefaultDomainPassword Policy	Edit the domain-linked account policy; do not assume an OU account-policy setting creates a departmental domain password policy.	Document domain policy ownership.
Kerberos authentication failure	Time skew or name/service issue	w32tm /query /status; klist; nltest /dsgetdc:ad.aster.example	Verify date/time, AD DNS and DC reachability; restore domain time hierarchy; retest.	Keep host/guest time consistent.
Duplicate SPN error	Service principal registered on more than one object	setspn -X; setspn -Q HOST/AST-FS01	Identify the legitimate service account before any deletion. For the baseline built-in host SPN, compare the computer account; escalate unfamiliar service SPNs rather than deleting all entries.	Never register copied SPNs without ownership checks.
Ping fails but web/share works	ICMP is blocked independently	Test-NetConnection AST-FS01 -Port 445	Inspect the specific ICMP rule if ping is required; preserve working service rules.	Validate each required protocol separately.
TCP check fails while DNS works	Service stopped or firewall blocks the endpoint	Get-Service; Test-NetConnection <target> -Port <actual port>	Check the task service locally, then named firewall rule profile/address/port; correct one setting and retest.	Avoid global firewall disablement.

Symptom	Cause	Diagnostic	Step-by-step repair	Prevention
PTR lookup fails	Reverse zone or PTR absent	nslookup 192.168.50.20	DNS Manager > Reverse Lookup Zones > New Zone > Primary AD-integrated > IPv4 > Network ID 192.168.50 > secure updates; add PTR host 20 to AST-FS01.ad.aster.example.	Treat reverse DNS as a documented diagnostic enhancement.
Backup share inaccessible	Wrong credentials or restrictive ACL	Test-NetConnection AST-BK01 -Port 445; net use	Reconnect using AST-BK01\backupwriter at the password prompt; verify both target ACLs.	Use a dedicated identity and restricted target.
Old recovery point disappears	Same remote destination reused	wbadmin get versions -backupTarget:<actual run path>	Use distinct documented run folders and verify each; do not claim a retention point that is absent.	Restore before retiring the older run.
Security file event absent	Audit subcategory or object SACL missing	auditpol /get /subcategory:"File System"	Enable the intended subcategory via GPO and add the small folder SACL; generate a fresh standard-user access and correlate it.	Validate logging end to end.
AD Sites console is wrong	dsa.msc opens users/computers	dssite.msc	Open dssite.msc for Sites and Services, dsa.msc for users/computers and gpmmc.msc for Group Policy.	Keep console names in the handover.
Additional DC not synchronised	DNS/RPC/SYSVOL or time problem	repadmin /replsummary; repadmin /showrepl	Restore AD DNS and network reachability; inspect Directory Service and DFS Replication logs before another promotion attempt.	Make replication health a prerequisite for failover.

P1-01 service-specific failures

Symptom	Cause	Repair
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
Backup share is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

P1-02 service-specific failures

Symptom	Cause	Repair
Backup target denied	Writer identity or ACL wrong	Retest the P1-01 SMB session; verify both share and NTFS permissions.
No backup version	Job incomplete or wrong subfolder queried	Use exactly Run01 and wait for Completed before enumerating versions.
Insufficient disk space	Backup destination is full	Check BK01 D: free space; preserve the previous verified backup before clearing any old run.

P1-03 service-specific failures

Symptom	Cause	Repair
Wrong backup date	Restore point selected before the test file existed	Choose the Run01 version created after seeding the file.
File seems missing	Restore preserves subfolder structure	Search C:\Recovered recursively for restore-demo.txt.
Hash mismatch	Wrong version or modified file	Do not declare pass; compare backup date and expected hash, then restore the correct recovery point.

P1-04 service-specific failures

Symptom	Cause	Repair
Access denied	Console is not elevated or account lacks administration rights	Sign out of the standard account. Sign in as ASTER\Administrator and right-click Windows PowerShell > Run as administrator.
Backup retention is unavailable	Role or service did not start	Open services.msc, locate the named service, inspect its status; use the task verification command and read the latest System event before starting it.
Remote check fails while local check succeeds	DNS, virtual NIC placement or a scoped firewall rule is wrong	Run ipconfig /all; confirm VMnet2 and 192.168.50.10 DNS; run Test-NetConnection to the specific service port before changing a firewall rule.

SECTION 10 OPERATIONS HANDOVER

Frequency	Owner	Action	Threshold and escalation
Daily	L1 operator	Check guest state, service status, DNS resolution, latest backup where present and new warning/error events	Any failed mandatory service or missed expected backup: record and escalate to trainer/L2
Weekly	L1 with reviewer	Repeat standard-user access sample, restore one disposable file where backup exists, inspect capacity and recent changes	Free guest/host disk below 20 percent: review growth; do not delete backups without retention review
Monthly	Trainer/L2	Review account/group ownership, maintenance plan, recovery timings and future platform requirements	Unowned account, persistent replication failure or unknown high-privilege grant: resolve before acceptance
Before maintenance	Change owner	Record baseline, intended reboot order, rollback and verification	Never reboot all redundant partners together for a single-node test
After maintenance	Change owner	Rerun task checks and record actual outputs	Close change only after acceptance tests pass

Patch approach: the institute supplies authorised updates suitable for the legacy lab before isolation. Record installed build/hotfixes. For a running exercise, back up accepted data, apply maintenance to one guest at a time, reboot, check service dependencies and then continue. This pack does not implement WSUS or claim current patch compliance.

Backup calendar for projects implementing WSB: the named operator starts the daily lab run while the VMs are on, verifies completion and retains only documented run folders. A proposed seven-day retention target is conditional on capacity; the executed lab proves only the actual stored and restored runs. Weekly restoration verifies usability. A powered-off laptop cannot meet a continuously operating service SLA.

Level	Role	When to escalate	Information to supply
L1	Trainee operator	A task verification fails after one documented targeted repair	Host, time, symptom, command output, latest change, affected users
L2	Trainer or assigned senior	Replication/authentication failure, recovery uncertainty or suspected broad exposure	L1 evidence plus baseline and rollback state

Level	Role	When to escalate	Information to supply
Owner	Training lead	Hardware/media unavailable or scope must change	Resource gap, effect on acceptance and proposed supported alternative

SECTION 11 RESUME AND INTERVIEW KIT

List this under Projects or Lab and Simulation Projects. Exact honest opening: "I built and tested this isolated Windows Server lab as part of my training. Aster is a fictional scenario; the configuration, troubleshooting and evidence are my own work." Do not list Aster as an employer or claim a retired Microsoft certification merely from course attendance.

STAR element	Resume-ready wording after completion
Situation and task	Built a self-directed new office infrastructure lab for a fictional services-company scenario using Windows Server 2016.
Action	Configured AD DS, DNS, DHCP, SMB and NTFS, using documented dependency checks and role-based access.
Result	Completed 4 specialist build tasks and five controlled failure/recovery drills, preserving actual verification evidence.
Result with measured values	Documented the real VM count, account count and measured recovery/validation times in the final handover; replace any template field with an observed value before using it.

11.2 Project description

Built an isolated Windows Server lab for a fictional services company. Implemented the assigned project services, documented dependencies and tested authorised and denied access. Diagnosed five controlled faults, verified recovery and prepared an operations handover. This self built simulation demonstrates practical administration; reported results describe only the machines, accounts, configurations and tests personally executed and evidenced during the training project.

11.3 Spoken project walkthrough

"My project was new office infrastructure for a fictional company called Aster. I started with the business problem: Anita in HR cannot retrieve onboarding files when a colleague is absent. I first built and verified the AD, DNS, DHCP and file-access foundation. Then I added the project services in dependency order. For each step I checked both the configuration and the user-visible result. I used standard accounts for access tests so administrator rights could not hide mistakes. I also injected five controlled faults one at a time, recorded the symptom and diagnosis, restored the intended state and retested. My evidence includes the actual hostnames, settings, command output and recovery observations. The main design limitation is that the VMs share one physical host. I describe this as a training simulation and can demonstrate the specific tasks I completed. If I extended it, I would address the documented failure-domain and platform requirements first."

11.4 Twenty five task-linked interview questions

Level	Task	Question	Model answer
Easy	B07	Why does AD depend on DNS?	The client uses DNS SRV records to locate domain services, not just A records for server IPs. I verified the LDAP DC-locator record.
Easy	B11	How is an OU different from a group?	An OU organises objects and scopes policy/delegation. A security group places a SID in access tokens for permissions.
Easy	B12	Explain the HR group nesting.	Anita and Vikas belong to GG_HR_Users, which belongs to DL_FS_HR_Modify; the resource group receives permissions.
Easy	B13	Does the dollar suffix secure a share?	No. It hides normal browsing; share and NTFS ACLs enforce access.
Medium	B13	Why test both an allowed and a denied user?	A folder working for its owner does not prove confidentiality; the denied case checks excess access.
Easy	B14	Which DHCP option supplies DNS?	Option 006. In the foundation it is 192.168.50.10; option 015 supplies ad.aster.example.
Medium	B14	Why is no router option configured?	The isolated foundation has one subnet and no router. Inventing a gateway would not provide connectivity.
Medium	B15	Can an OU GPO create a separate password policy for its domain users?	Not the ordinary domain account policy. This build uses domain scope; fine-grained policies would be a different feature.
Medium	B15	What proves that HR policy actually applied?	A gpresult report under Anita plus the observed Control Panel restriction and a Finance comparison.
Easy	B16	Why sign out after group membership changes?	A new logon refreshes the access token and avoids stale SMB sessions making the result misleading.
Medium	B01	Why is VMware DHCP disabled?	The Windows DHCP service must be the intended allocator; another responder can give clients wrong settings.
Hard	B07	Would a second VM guarantee availability?	No. VMs on one host share its power, storage and hypervisor failure domain.
Easy	P1-02	What is the backup source?	D:\Departments on FS01.
Easy	P1-01	Why use backupwriter?	It writes to a restricted target without becoming a local administrator.

Level	Task	Question	Model answer
Easy	P1-03	What is the difference between backup and replication?	Backup provides a recoverable historical state; replication synchronises changes including deletions.
Easy	P1-04	What does RPO 24 hours mean here?	A target for the age of recoverable data, conditional on successful daily runs; it is not proven by a schedule alone.
Medium	P1-03	What does the 30-minute RTO apply to?	One small test-file restoration, not a complete server or business recovery.
Medium	P1-03	Why restore to another location?	To inspect and compare the chosen version before overwriting current data.
Medium	P1-03	What is the hash comparison proving?	The restored file matches the recorded original bytes.
Medium	P1-04	Why use Run01 and Run02?	The same remote backup target can overwrite the earlier backup; separate targets make retention explicit.
Medium	P1-02	What if a backup job fails?	Preserve the last verified recovery point, inspect the error and rerun to an appropriate target; do not mark the schedule as completed.
Medium	P1-01	Can this survive laptop disk failure?	No, because source and target VMs share that disk.
Hard	P1-03	How do you measure restore time?	Record request/start and successful verification completion, then calculate elapsed minutes.
Hard	P1-01	What data should screenshots exclude?	Passwords, keys and unrelated personal information.
Hard	P1-04	What proves the project is finished?	All acceptance tests, evidence, five restored drills and handover sign-off, not just installed roles.

11.5 Eight gotcha questions

Question	Honest model answer
What would you do differently next time?	Separate production roles and failure domains, modernise the OS baseline and prioritise the measurable limitation specific to this project. Do not claim those changes were built.
What went wrong and how did you fix it?	Use one actual logbook incident: symptom, diagnostic command, root cause, exact fix and successful retest. Never invent an outage story.
Did you support 60 live employees?	No. Sixty is fictional business context. I implemented the documented lab accounts and VMs and can show their evidence.
Are screenshots sufficient?	No. I can reproduce the relevant test, explain the values and respond to a small live change.
Why not turn the firewall off?	It changes too many conditions and hides the actual service rule; I tested and repaired the specific path.
Can you claim production availability?	No. The host, storage and operational controls of this simulation are limited; I state the exact fault tested.
What if the expected output differs?	Check identity, machine, version and task prerequisites; record the actual output and resolve the difference before marking pass.
Is every command safe to paste at once?	No. Blocks span hosts and include intentional changes. I identify the machine and purpose, then run only the relevant task block.

SECTION 12 EXTENSION ROADMAP

Add-on	What it demonstrates	Indicative hours
Add a second DC	Project 4; authentication and DNS continuity	24
Controlled departmental storage	Project 2; quotas, namespace and replication	24
Off-host recovery target	A separate physical device and full server recovery plan	8

Extensions are planning items, not prerequisites secretly required to pass this project. Teach any new dependency before assigning it. A new platform, certificate authority or physical failure domain needs its own build sheet and acceptance tests.

Sources and authoring validation

Training scope: the attached MCSA_CPR_with_Task_Lab_Work.xlsx, worksheet CPR Tracker, supplied with this request. Delivery requirements: the attached master prompt. The workbook records assigned/covered work, not per-trainee competence; the acceptance checks establish execution evidence.

ID	Reference	Purpose
S01	Microsoft Windows Server 2016 Evaluation Center	https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2016 Evaluation media and time-limited use; obtain lawful client media separately.
S02	Microsoft wbadmin start backup	https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/wbadmin-start-backup Remote target behaviour and backup parameters.
S03	Microsoft DFS Replication overview	https://learn.microsoft.com/en-us/windows-server/storage/dfs-replication/dfs-replication-overview DFS Replication concepts and relationship to namespaces.
S04	Microsoft DFS Replication FAQ	https://learn.microsoft.com/en-us/windows-server/storage/dfs-replication/dfs-replication-faq Closed files, conflicts, quotas and replication limitations.
S05	Microsoft WDS boot image support	https://learn.microsoft.com/en-us/windows/deployment/wds-boot-support Version-specific deployment-media limitations.
S06	Microsoft DHCP failover overview	https://learn.microsoft.com/en-us/windows-server/networking/technologies/dhcp/dhcp-failover Lease/configuration replication, modes and partner-state behaviour.
S07	Microsoft Add DhcpServerV4Failover	https://learn.microsoft.com/en-us/powershell/module/dhcpserver/add-dhcpserverv4failover Failover command parameters; check local help for installed version.
S08	Microsoft wdsutil set server	https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/wdsutil-set-server PXE response and pending-client approval settings.

Authoring review checks file integrity, required sections, task dependencies, evidence identifiers, diagram assets and document layout. Runtime validation on Windows Server 2016 remains the trainee/institute pilot requirement. Illustrative outputs and blank report fields are clearly separated from actual results. This pack contains no fabricated Windows screenshots.